

General Industries Hub

Enterprise Inventory Management

Product Manual & User Catalogue

Version 1.2.0 - Generated 2026-08-06

Confidential - for authorized personnel only

GI Hub - Multi-Site Inventory, Procurement & Asset Management

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General Industries Hub - Product Manual & User Catalogue

Version 3.0 - Multi-Site Warehouse Inventory ERP + Procurement Chain Document Scope: Complete operational reference for every role, page, tab, and element built into the system. Companion documents: the Standard Operating Procedure, for day-to-day procedure organised by task rather than by screen; and the architecture reference, for whoever administers the server.

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1. Introduction & System Overview

1.1 What the system does

The General Industries (GI) Hub is a multi-site warehouse inventory, procurement and asset-management system. It tracks every material movement - issues, receipts, returns and corrections - through a two-stage approval process, and gives you live stock figures, oldest-stock-first discipline on anything with an expiry date, stock valuation, serialised tool and equipment tracking, and a complete audit trail across every site you manage.

It runs in a web browser on a phone, a tablet or a desktop, and is also available as an installable app for Windows, macOS and Android.

1.2 Core principles

Principle	What it means
Stock is calculated, never stored	Stock on hand is always received, minus consumed, minus returned, recalculated from the movement history every time it is shown. No running total is kept anywhere, so no
Two-stage approval	Store Keepers stage entries step-HOD ready and commits at End-of-Day. Nothing touches the permanent ledger without HOD approval.
Site isolation	HODs and Supervisors see only their own site's stock. Only Admin sees all sites. Cross-site moves require formal
Audit-first	Every step approval action is recorded with the user's name, the exact time and what changed. Even a deletion leaves a trace.
Nothing to maintain	Upgrades apply themselves when the system starts. Nobody has to run a database update by hand.
Procurement chain	One continuous workflow from a site's request through to material on the shelf. Logistics owns purchase orders; Warehouse owns physical receiving and delivery notes; the Site HOD approves what is being sent; the Store Keeper
RL/BL strict separation	Rubber Lining and Brick Lining items NEVER share a PO line group, a DN, or a warehouse aggregation. The system rejects mixed-family DNs by design and tags each line with its family on insert.

Warehouse-blind pricing	Warehouse users can see materials and quantities but NEVER see Unit_Price, Total_Price, or any monetary header field on a PO. Three independent enforcement layers guarantee this.
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1.3 The transaction lifecycle (the heart of the system)

Every stock movement travels the same four steps. Nothing reaches the permanent record until a Head of Department has approved it.

Step	Who does it	What happens	Where it sits afterwards
1	Store Keeper	Enters the issue, receipt or return on the Entry Log	Staged, awaiting approval
2	Head of Department	Reviews it at End-of-Day and may edit the quantity, approve, or reject with a reason	Still staged until they decide
3	System	Commits the approved entry to the permanent stock	Permanent - stock figures move now
4	System	Sends Sends the WhatsApp and in-app notifications, and writes the audit entry	Permanent audit history

Receipts, issues, returns and stock adjustments all follow this same shape. A rejected entry never reaches step 3, so it never affects a stock figure - but the rejection itself is still recorded, together with the reason.

1.3a The procurement chain

Buying material is a relay. Each role does one thing and hands over to the next, and every handover appears in the receiving role's own queue.

Step	Role	Action	Lands in
1	Site HOD	Submits a Purchase Request	Logistics - Incoming PRs
2	Logistics	Issues the Purchase Order	Site HOD is notified; Admin sees it for oversight
3	Logistics	Assigns the order (or individual lines) to a warehouse	Warehouse - Incoming Assignments
4	Warehouse	Acknowledges, then records the physical arrival from	Warehouse - Receiving
5	Warehouse	Prepares the Delivery Note	Logistics - DN approval queue
6	Logistics	Approves the delivery date	Site HOD - DN Approvals

Step	Role	Action	Lands in
7	Site HOD	Approves the contents	Store Keeper - Incoming DNs
8	Store Keeper	Confirms physical arrival on site	Permanent stock record; the delivery closes

Three side-paths exist for when things do not go to plan:

- Vendor returns - any role can raise one. The order line reopens so the material can be re-delivered.
- Reschedules - Warehouse or Site HOD request a new date; Logistics decides.
- Force-closure - Logistics only, for an order that will never complete. Both Admin and the Site HOD are notified automatically.

A Delivery Note may never mix Rubber Lining and Brick Lining material. The system refuses to create one, rather than warning about it.

1.4 Currency, dates, units

- Currency: Saudi Riyal (SAR) throughout.
- Dates: shown as DD/MM/YYYY or DD MMM YYYY.
- Time zone: the site's local time. Audit entries record the time to the second.
- Units of measure: each item carries its own unit - pieces, boxes, rolls, cans, metres, kilograms. There is no automatic conversion: an item is issued in the same unit it was received in. If a material arrives in boxes and is used in pieces, record the conversion in the item's description and keep one unit for both directions.

2. Roles, Permissions & Page Access

2.1 Role hierarchy

Roles are ranked, but the ranking is not a straight line - two pairs sit side by side at the same level and differ only in what they are scoped to.

Level	Role	Scoped to
0	Store Keeper	One site
1	Warehouse User	One warehouse
1	Supervisor	One site
2	Head of Department	One site
3	Logistics	All sites
3	Auditor	All sites, read-only
4	Admin	Everything

A higher rank does not open a lower role's workspace. The Logistics Portal, Warehouse Portal, Entry Log, HOD Portal, Supervisor Portal, Material Estimator and Man-Hours pages are locked to their own role exactly - Logistics outranks a Head of Department but still cannot open the HOD Portal. Admin is the single exception and reaches every workspace deliberately, for support.

2.2 Page access matrix

Page	Store Keeper	Warehouse User	Supervisor	HOD	Logistics	Auditor	Admin
Live Dashboard	[X]	[OK]	[OK]	[OK]	[OK]	[OK]	[OK]
Entry Log	[OK]	[X]	[X]	[X]	[X]	[X]	[X]
HOD Portal	[X]	[X]	[X]	[OK]	[X]	[X]	[X] (hidden - uses Admin)
Logistics Portal	[X]	[X]	[X]	[X]	[OK]	[X]	[OK] (admin)
Warehouse Portal (NEW)	[X]	[OK]	[X]	[X]	[X]	[X]	[OK] (admin shadow, picks WH)
Admin	[X]	[X]	[X]	[X]	[X]	[X]	[OK]
Reports	[X]	[OK]	[OK]	[OK]	[OK]	[OK] (read + download only)	[OK]

Records (Inventory, ledgers, POs, PRs)	[X]	[OK]	[OK]	[OK]	[OK]	[OK]	[OK]
Supervisor Portal	[X]	[X]	[OK]	[X]	[X]	[X]	[OK] (admin
Material Estimator	[X]	[X]	[X]	[OK]	[X]	[X]	flow) [OK]
Man-Hours (NEW)	[X]	[X]	[X]	[OK]	[X]	[X]	[OK]
Auditor view	[X]	[X]	[X]	[X]	[X]	[OK]	[OK]

> Exact-role locks: Entry Log, HOD Portal, Supervisor Portal, Logistics Portal, Warehouse Portal, Material Estimator and Man-Hours are locked to their own role - a higher-ranked role does not inherit access. Admin reaches them for support purposes.

2.3 Site scope by role

Role	What they see
Store Keeper	Their own site only - they cannot view another site's stock.
Warehouse User (NEW)	Their own warehouse only - POs assigned to them, DNs they've prepared, items received from vendors. Tied to users.Warehouse_ID.
Supervisor	Their own site only - Reports, Live Dashboard, Burn Rate all site-locked.
HOD	Their own site only - but they can REQUEST material from
Logistics (NEW)	All sites globally for PRs and POs they manage. No site lock - they sit above the site boundary.
Auditor (NEW)	All sites globally, read-only. Sits at level 3 so it is not site-locked - an auditor pinned to one site could not audit. It
Admin	All sites + all warehouses globally - has the "All Sites" filter on every multi-site view; warehouse picker in sidebar

when

shadowing the Warehouse Portal.

2.4 Default seeded accounts

The first time the app starts, these accounts are created. Change the passwords immediately.

Username	Password	Role
admin	admin2026	Admin
hod	hod2026	HOD

Username	Password	Role
supervisor	super2026	Supervisor
worker	floor2026	Store Keeper

No default logistics or warehouse_user accounts are seeded. Both roles are strictly admin-created - go to Admin Portal -> Users -> Add user. This is intentional: the procurement chain has commercial visibility (Logistics sees prices, Warehouse routes inventory) and seeded credentials would be a security liability. When you create a warehouse_user, set their Warehouse_ID to one of the values from Admin Portal -> Master DB Editor -> warehouses table - without it, the user lands on the Warehouse Portal and sees an error telling them to ask Admin.

3. Login, Sidebar & Common Elements

3.1 Login screen

Elements:

- Username text box - your assigned username (case-sensitive).
- Password text box - masked input. Min 8 characters policy enforced on creation.
- Sign In button - checks your credentials and takes you straight to the first page your role can open.
- "Don't have an account? Request access" link - opens a self-service registration form.

Registration form elements:

- Username - must be unique
- Password / Confirm Password
- Role requested - selectbox: Store Keeper / Supervisor / HOD (Admin role cannot be self-requested)
- Site_ID - which site to associate the account with
- Phone Number - used for WhatsApp alerts (format: +966 5X XXX XXXX)
- Submit Request button - creates a pending_users row + audit log + WhatsApp alert to all admins

Admin must approve via Admin Portal -> Users tab before the user can log in.

3.2 Sidebar (visible after login)

Every page shares this sidebar. Reading top to bottom:

Element	Purpose	Notes
GI Hub bolt icon + version	Branding	Shows v3.0.0
Role card	Your username + role badge	Color-coded: grey=Store Keeper, emerald=Warehouse, blue=Supervisor, indigo=HOD, sky=Logistics, gold=Admin
Notifications bell (NEW v3.0)	Unread count + inbox dialog	Red badge if N>0, primary button changes to "Open inbox (N unread)". Modal shows recent procurement events with mark-read controls. See §3.6
"Navigate to:" radio	Page picker	Only shows pages allowed by your role
INVENTORY ALERTS section	Compact stock-alert badge	Visible to Supervisor/HOD/Admin only. Shows count of items below
Bug/Feature reporting	Opens a dialog	See §3.4
Theme toggle	Dark / Light mode switch	Persists per-session

Element	Purpose	Notes
Sign Out button	Ends the session	Audit-logs the LOGOUT event

3.3 Brand header (top of every page)

Variant	Used by	Visual
render_brand_header	Live Dashboard, Entry Log	Gold subtitle accent + current date
render_brand_header_hod	HOD Portal	Purple subtitle accent + current date
render_brand_header_admin	Admin Portal	Gold accent + green/amber pulse chip ("All systems operational" / "Degraded") + current date

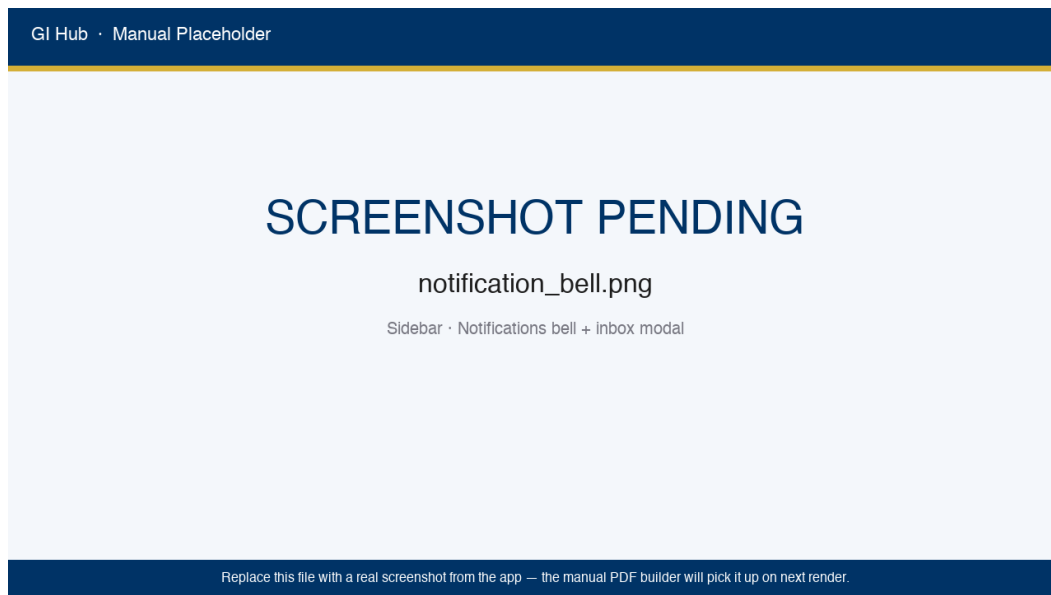
3.4 Bug / Feature reporting dialog

Available to every user from the sidebar:

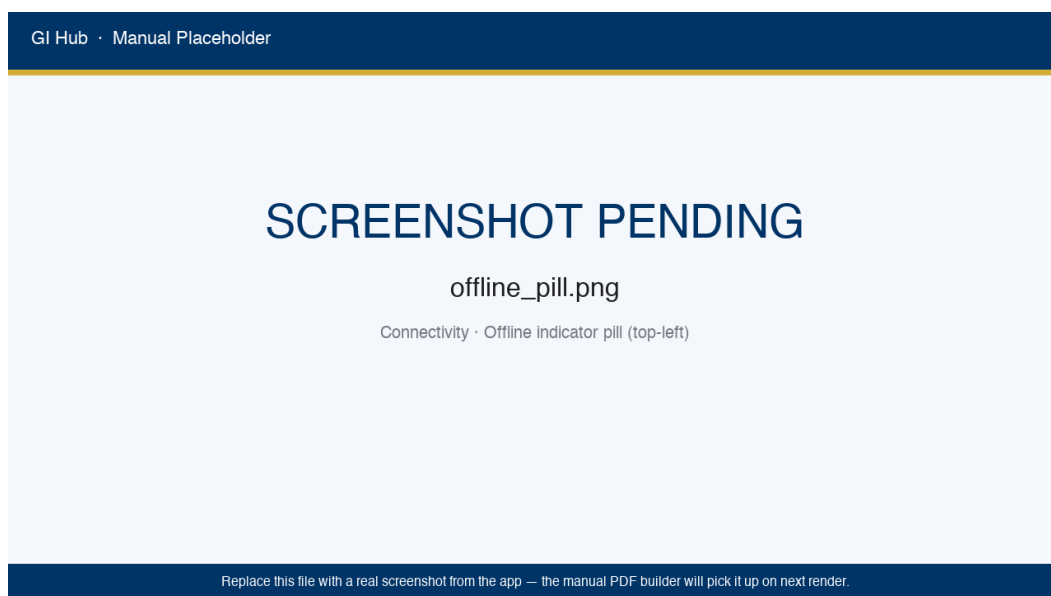
- Type selectbox - Bug Report / Feature Request
- Page dropdown - which page the issue/idea relates to (Live Dashboard, Entry Log, HOD Portal, Admin Portal, Reports, Other)
- Description textarea - up to 200 characters
- Submit button - writes to bug_reports table with user, timestamp, type, page, description. Visible to Admin in the Admin Portal Reports & Bugs tab for triage.

3.6 Notifications bell (NEW in v3.0)

Sits between the role card and the navigation radio. Every signed-in user sees their own personalised inbox of procurement-chain events.



Sidebar - Notifications bell with unread badge + inbox modal



Connectivity - offline indicator pill appears top-left when the browser loses network (Phase 7E)

The button:

- No unread: "Open inbox" (secondary button, no badge)
- 1+ unread: "Open inbox (N unread)" (primary button, red pill badge with the count, capped at "99+")

The inbox dialog (click the button):

- Only unread toggle - on by default. Flip off to see your full history.
- Per-notification card - colour-coded left border by severity:
 - Critical (red) - force-closures, T-0 delivery reminders
 - Warning (amber) - T-1/T-2 reminders, reschedule requests
 - Success (green) - DN approved / received successfully
 - Info (blue) - PR submitted, PO issued, assignments
- Per-row Mark read - flips just that notification
- Bulk [OK] Mark all as read - flips every visible row (respecting the role + site + warehouse scope)

you'd see normally)

What gets sent here:

Event you'll see	Triggered by
New PR from a site (Logistics only)	Site HOD pressing " Submit PR(s) to Logistics"
PO issued (Site HOD)	Logistics creating a PO against your site's PR
PO assigned to warehouse (Warehouse only)	Logistics routing a PO to your WH
DN awaiting your approval (HOD)	Logistics approving a DN delivery date
Incoming DN ready to receive (SK)	HOD approving DN content
Reschedule requested (Logistics)	Warehouse or HOD asking to push a date
Reschedule decided (requester)	Logistics approve/reject
Force-closure (Admin + originating HOD)	Logistics force-closing a PR/PO/line
Vendor return raised (Logistics)	Any role raising a return
Delivery reminder T-2 / T-1 / T-0	The daily sweep job (see §9)

The bell is tolerant - if a notification helper errors, the badge silently shows 0 instead of crashing the sidebar. In-app notifications ALWAYS fire; the WhatsApp side is gated by toggles in config.WHATSAPP_TRIGGERS (see §9).

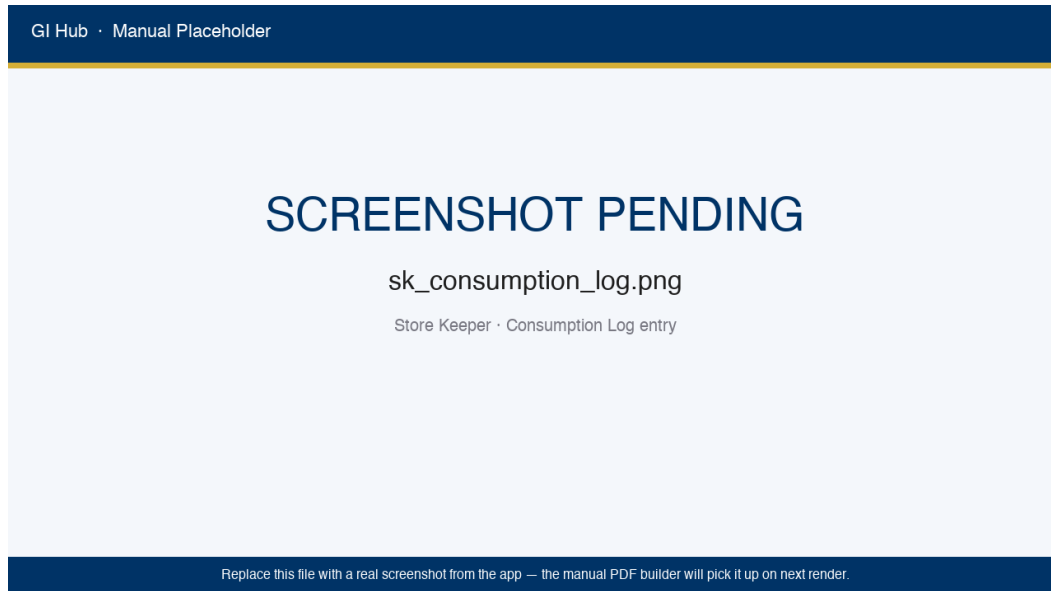
3.5 Overdue Returnable banner (Store Keepers only)

When a Store Keeper logs in and there are overdue returnable items at their site, a red banner appears at the top of any page they navigate to:

> [!] OVERDUE ITEMS - Action Required: N borrowed item(s) past expected return: <item names>. Go to the Returnable Items tab to follow up.

4. Store Keeper Manual

The Store Keeper is the warehouse-floor operator. They see only the Entry Log page and the sidebar shell.



Store Keeper Consumption Log entry - main daily workflow

4.1 Pages visible

- Entry Log
- Locator - find which rack a material is on, or scan a rack to see what should be on it (see §21.8)
- Assets - the serialised tools and equipment register, with its own record and location history for each physical item (see §21.7)

The Locator and Assets pages are available to every role, Store Keeper included. They are deliberately placed at the top of the menu rather than buried inside a portal, because the people who need them are usually standing in the warehouse holding a scanner.

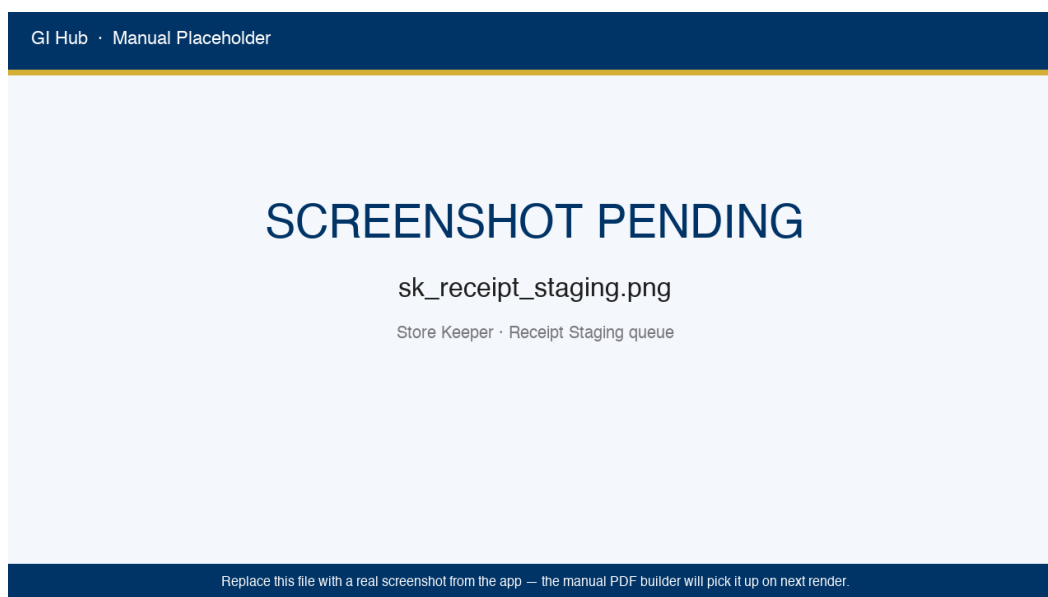
4.2 Entry Log - Tab structure

The Entry Log has four tabs:

1. Consumption Log - record material issued out
2. Receipt Staging - record material arriving in
3. Returnable Items - track tools/items temporarily issued
4. Stock Count - submit physical-count reconciliations

4.3 Entry Log -> Consumption Log

This is where every material consumed by site operations is recorded.



Receipt Staging queue - review incoming materials before submitting to HOD



Supervisor Requests tab - approve / reject worker material requests

4.3.1 Top section: Bulk OCR upload (expander)

Upload Handwritten Consumption List (OCR) - for when you have a handwritten list to bulk-stage instead of typing each row.

Element	Purpose
Input method radio	Switch between "Image upload (vision AI)" and "Paste text"
File uploader	Upload PNG/JPG/JPEG/WEBP of handwritten list (vision model reads it)
Extract rows from image button	Triggers vision AI extraction -> preview rows
Paste text area	Alternative - paste tab/comma/pipe-separated rows
Parse pasted rows button	Parse the textarea

Element	Purpose
OCR Review grid	Editable preview of extracted rows with ambiguous-match pickers
Confirm All & Stage button	Pushes all reviewed rows into pending_issues

4.3.2 Mobile camera barcode/QR scanner (expander)

Barcode / QR Scanner - opens the browser camera and reads SAP codes off labels.

Element	Purpose
Camera feed	Live preview (requires camera permission)
Manual SAP code entry text box	Fallback if camera unavailable - also accepts pasted scanner output
Detected code display	Shows the last successfully scanned code in green

When a code is scanned, the material selectbox auto-populates so you don't have to search.

4.3.3 Scan / Add New Item to Queue (main expander, expanded by default)

This is the workhorse panel. Reading top to bottom:

A. Recently-scanned ring buffer

Element	Purpose
Recent: pills	Up to 5 quick-tap buttons showing your last 5 items (SAP code + truncated description). Tapping one auto-fills the material selectbox.

B. 1. Select Material

Element	Purpose
Search by SAP Code or Description selectbox	Type-ahead search across the inventory master. Format: [SAP_Code] Description.

When you pick a material, two cards appear automatically:

C. Item Snapshot card

A compact dark card showing:

- SAP code (gold monospace)
- Material description (bold)
- 3-stat strip:
- SITE STOCK (color-coded: red=empty, amber=below min, green=ok)
- 30-DAY BURN (units consumed last 30 days)
- DAILY RATE
- Inline 70x22 SVG sparkline - last 30 days consumption trend
- Status badge bottom-right (OK / Low / Below Min / Empty)

D. FEFO panel (lot suggestion)

If lots exist for this item at your site:

Element	Purpose
FEFO - First Expiry, First Out header (amber)	Banner
Per-lot row	Lot number (monospace) - xqty (dim) - "Exp <date>" (amber if <90d, red if expired) - USE FIRST badge on the first row

The system will automatically attach the top-FEFO lot number to your consumption row when you submit.

E. Pull from a different lot (FEFO override) - expander

Only renders when 2 or more open lots exist for this item at your site.

Element	Purpose
Lot to pull from instead selectbox	Default: "Keep FEFO suggestion". Other options: all other open lots with expiry + remaining qty shown
Reason for override text box (200 char max)	Mandatory. Min 5 characters to activate the override
Amber confirmation banner	Appears once override is active: "FEFO override active: pulling from <chosen> instead of <suggested>"

When activated:

- Your consumption row gets Lot_Number = chosen + FEFO_Override = reason
- Audit log entry FEFO_OVERRIDE is written with full context
- A WhatsApp alert is queued to the site HOD in real time

F. 2. Fill Entry Details (dynamic form)

Fields appear based on the pending_issues schema. Every required field is marked with *.

Field	Type	Purpose
Date	Date picker (default: today)	When the consumption happened
Quantity	Number input (min 0.1)	How much you're issuing. Above this field is a stock badge showing current site stock; a red [!] warning appears if you type more than available.
Work_Type	Selectbox (sourced from system_settings.Work_Type)	Classification: PM Work, Breakdown, Project, Shutdown, Inspection, etc.
Issued_By	Text (auto-filled with your username)	Who handed it over
Issued_To	Text (smart-defaults from your last entry)	Recipient personnel/department
Tank_No	Text	Equipment tag if applicable
Serial_No	Text (optional)	Item serial number if tracked
PR_Number	Text (smart-default)	Linked purchase request if applicable
Remarks	Text (optional)	Free-form notes

Hidden side-effects:

- Site_ID is auto-set to your site

1

Lot_Number is auto-set to FEFO top suggestion (or your override)

- status is set to 'draft' until you submit

G. **[!] Override Expiry Warning (amber card with checkbox)**

If the system detects a short-dated batch at your site for this item, it will hard-block the Add to Grid button until you check this box, confirming you've physically pulled from the expiring batch first.

H. **Add to Grid button**

Validates that:

- A material is selected
- No required fields are empty
- Quantity <= current site stock (over-issue guard)
- Expiry override is checked when needed

On success: inserts a pending_issues row with status='draft', toasts [OK] Added to staging queue, reruns the page.

4.3.4 Staging Queue section

Element	Purpose
Staging Queue header + count badge	Shows how many draft rows you've accumulated
Data editor table	Edit any row inline before submitting
Save Draft Edits button	Persists edits in-place (rows stay as drafts)
Submit Grid to HOD button	Flips all draft rows to status='pending_hod' + queues a WhatsApp alert to the site HOD with the full item list

After submission, the rows are locked from your view - they appear in HOD Portal -> EOD Commit tab for approval.

4.4 Entry Log -> Receipt Staging

For when materials arrive at your site.

4.4.1 Upload Delivery Note (OCR) - expander

Identical lane structure to consumption OCR - upload an image of the delivery note OR paste text. Vision AI extracts rows for bulk staging.

4.4.2 Add Receipt to Queue - expander (expanded by default)

A. 1. **Select Material**

Element	Purpose
Search by SAP Code or Description selectbox	Same type-ahead search as consumption

When picked, a blue-tinted info card shows:

—

Mat Code

- UOM

B. 2. Fill Receipt Details (dynamic form)

Field	Type	Purpose
Date	Date picker (today)	Delivery date
Quantity	Number input	Units received
Supplier	Text (optional)	Vendor name
Lot_Number	Text (optional)	If blank + Expiry_Date is set, system auto-generates LOT-YYYYMMDD-SAP
Expiry_Date	Date picker (optional)	Lot expiry - when set, triggers lot master entry
PR_Number	Text (optional)	Link to an open Purchase Request
Serial_No / Vehicle_No / DN_No / Pallet_No / Mob_From / Mob_To / Prepared_by / Driver_Name	Text (optional)	Logistics tracking fields
Remarks	Text	Free notes

C. Add to Receipt Queue button

Validates: material picked, required fields populated. Inserts a pending_receipts row with status='draft'.

4.4.0 Incoming Delivery Notes from Warehouse (NEW in v3.0)

A new expander appears at the TOP of the Receipt Staging tab. It's only populated when the Warehouse has prepared a DN bound for your site AND your HOD has approved it. If empty, the expander shows: "Nothing inbound. Logistics -> HOD-approved DNs will appear here for you to confirm physical receipt."

When DNs are inbound, each appears as its own container:

Element	Purpose
DN header line	DN <number> - PO <number> - Warehouse <id> - DN Date
Line count + total qty	Alert: how many SKUs, how many units total
View lines expander	Material_Code, Description, Qty, UOM, Lot_Number, Expiry_Date, Remarks - read-only preview of every line
[OK] Mark as Received button	Confirms physical receipt. One receipt is recorded per delivery line, each keeping its delivery note, warehouse and originating order so the trail is complete; the delivery closes

When to use:

- The Warehouse delivered the materials physically to your site
- HOD already approved the DN content (you'll see it appear without any action)
- You've inspected the goods and they match the DN qty + lot

When NOT to use:

- Direct deliveries from supplier to your site (those go through the existing Add Receipt to Queue -> Submit to HOD flow below; the email-driven PR/PO path remains supported)
- Partial receipts (the current flow assumes you confirm the full DN qty - for partials, raise a Vendor Return on the diff after confirming and ask your HOD)

This is the FINAL step in the procurement chain. After you click Mark as Received:

- Logistics sees dn_received_by_sk in their notifications
- Warehouse sees the same
- The PO and PR move toward closure if this DN completes the order

4.4.3 Receipt Draft Queue section

Element	Purpose
Receipt Draft Queue header + count badge	Number of drafts you've accumulated
Data editor table	Inline edit before submit
Save Draft Edits button	Persists
Submit to HOD for Approval button	Flips drafts to status='pending_hod' + WhatsApp alert to HOD with item list

4.5 Entry Log -> Returnable Items

For tools, gauges, fittings temporarily handed to personnel - items that should come back.

4.5.0 Smart Scan workflow (new - 2026-06)

A two-step camera flow sits at the top of the tab. Use it to skip typing borrower + tool details by hand; the existing manual form still works exactly as before and stays visible underneath.

Step 1 - Scan the employee badge

1. Click into the Smart Scan (Beta) expander.
2. Hold the borrower's printed badge in front of the camera. (Badges are generated in Admin Portal -> Employees -> Roster + Badges - see §7.6.)
3. The system decodes the QR (which carries only the ID_Number, no PII), looks it up in the Employees master, and shows:
 - [OK] Green success card - EMP-1042 - Ahmed Ali (+9665...) - the borrower's name + phone are now staged.
 - [X] Red error card - Badge couldn't be read OR the employee is inactive/suspended. Re-snap or use the manual form.

Step 2 - Scan the tool

1. After a successful badge scan, a second camera input appears. Point it at the tool.
2. The active YOLO model (managed in Admin Portal -> Tool Catalogue, see also docs/cv_training_guide.md for how the model gets trained and promoted) classifies the image. Three

outcomes:

- Confidence ≥ 0.75 -> [OK] Auto-fill. The detected tool name + borrower details flow into the manual form below; review and click "Issue Item".
- Confidence 0.30-0.74 -> [!] A "Top candidates" radio appears. Pick the right one and click "Use this tool". The form pre-fills.
- No active CV model OR confidence < 0.30 -> An info banner explains the fallback. Borrower fields stay pre-filled; type the tool name into the manual form yourself.

Click "Start over (clear scan)" any time to reset and scan a different borrower.

What happens if the AI is wrong

You're never locked in. Every Smart Scan write-through goes into the manual form's fields - Name, Phone, Material name. Edit anything that's wrong before clicking "Issue Item". The submitted record stores `cv_detected=1` and `cv_confidence` so admin reports can later show adoption + accuracy telemetry. Manually-corrected loans don't get tagged as CV - that's intentional, see §10 for the data model details.

Automatic WhatsApp reminders (Phase 6E)

Once a loan is in the system, the WhatsApp worker fires four reminders automatically. The cadence and recipient list is:

Offset	Severity	In-app	WhatsApp
T-2h (2 hours before due)	info	Site SK badge	Borrower
T-0 (due now)	warning	Site SK badge	Borrower
T+2h (2 hours overdue)	warning	Site SK badge	Borrower + every Site SK
T+24h (24 hours overdue)	critical	Site SK + Supervisor badges	Borrower + every Site SK + every Site Supervisor

The borrower's phone is resolved first from the Employees master (CV-scanned loans) and falls back to the manually-typed `borrower_phone` column for legacy/manual loans. The system de-dupes per loan + offset so restarting the worker mid-day never causes double-fires. Admins can mute any of the four events via `config.WHATSAPP_TRIGGERS` if a particular escalation gets too noisy.

4.5.1 Issue a Returnable Item - expander (expanded by default)

Field	Type	Purpose
Material / Tool Name	Text	What you're handing out
UOM	Text (e.g., Pcs, Set)	Unit
Quantity	Number (min 0.1)	How many
Borrower Name	Text	Person taking custody
Borrower WhatsApp Number	Text (optional, +966 ...)	For overdue alerts
Expected Return Date	Date (default: tomorrow)	When you expect it back
Expected Return Time	Selectbox: 04:15 PM, 06:15 PM, Custom Time...	Time-of-day expectation
Issue Item button	Validates name+borrower, writes to <code>returnable_items</code>	

4.5.2 Currently Borrowed Items section

Shows all borrowed (not yet returned) items at your site.

- Overdue red banner - when one or more items are past their expected return time
- Borrowed items table (styled HTML) with columns: ID, Material, UOM, Qty, Borrower, Phone, Given Time, Expected Return, Status (pill: green "On Loan" / red "[!] Overdue")
- Mark as Returned section:
- Selectbox of borrowed items
- [OK] Mark as Returned button - updates status='returned', busts cache, toast confirms

4.6 Entry Log -> Stock Count

For reconciling physical shelf count with system stock when they don't match (damage, expiry disposal, miscount, found-extra).

4.6.1 1. Select Material to Count

Element	Purpose
Search by SAP Code or Description selectbox	Pick the material to reconcile
Help caption	Explains when to use this tab

4.6.2 2. Enter Count Details (only renders after picking a material)

Left column

Element	Purpose
System Qty card	Read-only dashboard showing current stock at your site (auto-fetched)
Counted Qty input	What you actually count on the shelf right now (defaults to system qty)
Variance preview banner	Color-coded: green " Found N extra" or red " Short by N" or grey "No variance"

Right column

Element	Purpose
Reason Code selectbox	9 options. Default depends on variance direction.
Notes textarea (300 char max)	Free explanation: "found behind shelf 3", "damaged in transit", etc.

Reason codes:

Code	Label
cycle_count	Cycle count correction
damaged	Damaged / unusable
expired_disposal	Expired - disposed

Code	Label
miscount_in	Miscount - found extra
miscount_out	Miscount - short
lost	Lost / unaccounted
theft	Suspected theft
return_to_supplier	<- Returned to supplier
other	Other (see notes)

4.6.3 Submit section

Element	Purpose
Amber warning banner	"Submitting this sends the count to your HOD for approval.
Submit Count for HOD Approval button	Disabled while generating the approval process: writes to stock_adjustments with status='pending_hod', queues WhatsApp to site HOD with count details, audit-logs.

4.6.4 Recent Adjustments at This Site (history)

Read-only table showing: ID, SAP Code, Material Name, Variance, Reason, Status, Submitted By, Submitted At.

4.7 Store Keeper - Use Cases

Use Case 1: Issue 10 Pipe Gaskets to PM Work

1. Log in -> Entry Log -> Consumption Log tab
2. Open Scan / Add New Item to Queue
3. Search "Pipe Gasket" -> pick from dropdown
4. Read the stock badge (e.g., " 156 Pcs"), confirm FEFO lot suggestion
5. Quantity: 10
6. Work_Type: PM Work
7. Issued_To: name of recipient
8. Tank_No / PR_Number: as applicable
9. Click Add to Grid
10. Repeat for any other items in this batch
11. Review the Staging Queue table -> edit if needed -> click Submit Grid to HOD

The row(s) are now in HOD's EOD queue. Stock won't decrease until HOD commits EOD.

Use Case 2: Receive a delivery of 200 units with expiry date

1. Entry Log -> Receipt Staging tab
2. Open Add Receipt to Queue
3. Pick material, fill Quantity (200), Supplier, Expiry_Date (e.g., 2027-06-11)

4.

Lot_Number: leave blank (system will generate LOT-20260611-<SAP>) OR type the supplier's lot ID

5. PR_Number: link if applicable
6. Click Add to Receipt Queue
7. Click Submit to HOD for Approval when done staging

Use Case 3: Found 5 extra units of an item (physical count > system)

1. Entry Log -> Stock Count tab
2. Pick the item from the dropdown
3. Read the System Qty card (e.g., "95")
4. Counted Qty: 100
5. The variance banner turns green: "Found 5 extra"
6. Reason: Miscount - found extra
7. Notes: e.g., "Found in box behind shelf 3"
8. Click Submit Count for HOD Approval

HOD will approve in HOD Portal -> Adjustments. After approval, your live stock matches reality and a synthetic receipt row of +5 is posted to the ledger.

Use Case 4: Issue a tool to a borrower

1. Entry Log -> Returnable Items tab
2. Issue a Returnable Item
3. Material: Torque Wrench 1/2", UOM: Pcs, Qty: 1
4. Borrower Name, WhatsApp Number, Expected Return: today 06:15 PM
5. Click Issue Item

If the item is not returned by 06:15 PM, the borrower (and you) get a WhatsApp alert.

Use Case 5: FEFO override (the right bin is blocked)

1. Entry Log -> Consumption Log -> pick material with 2+ open lots
2. Scroll to FEFO panel - note the suggested lot (earliest expiry)
3. Open Pull from a different lot expander
4. Pick a different lot from the dropdown
5. Type a reason >= 5 chars (e.g., "FEFO bin blocked by pallet, clearing 1700")
6. Amber confirmation banner appears
7. Fill quantity + other fields
8. Click Add to Grid

Your HOD gets a real-time WhatsApp alert documenting the override.

4.8 Store Keeper - FAQ

Q: I can't see Live Dashboard, HOD Portal, or Reports. Did I lose access? A: No - Store Keepers only have access to Entry Log by design. If you need to view stock, ask your Supervisor or HOD.

Q: I submitted to HOD by mistake. Can I cancel? A: Not directly. Contact your HOD - they can reject the row in their EOD Commit review.

Q: The "Add to Grid" button is disabled or the form rejects my entry. A: Check three things: (a) is quantity > current stock? (b) is the expiry override needed but unchecked? (c) are all required * fields filled?

Q: I tried to issue 20 but the system says only 15 available. A: Believe the system. Either physically recount, or if there is genuinely 20 and the system disagrees, use the Stock Count tab to log a +5 adjustment.

Q: I don't have a barcode scanner. How do I quick-find items? A: Type a partial SAP code or description in the selectbox - it has type-ahead search. Or use the Recently-scanned pills if you've used the item recently.

Q: WhatsApp alerts aren't reaching the HOD when I submit. A: HOD's phone number must be set in their user profile. Tell the HOD to update it via Admin (or check with Admin directly).

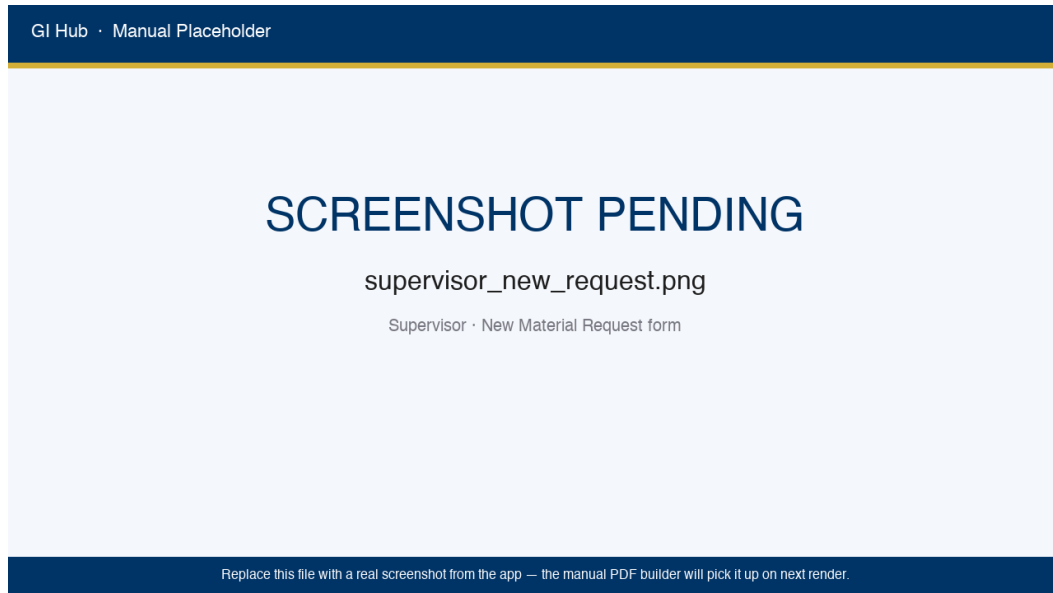
Q: My borrowed item is overdue but I haven't received an alert. A: WhatsApp alerts are sent by a background service. Ask your Admin to confirm it is running - queued messages send as soon as it is back, so nothing is lost in the meantime.

Q: I uploaded an OCR image and it extracted wrong items. A: The OCR review grid lets you fix every row before staging. Pick the correct material from the suggested-matches dropdown, or type it manually.

Q: How do I see my recent activity? A: Submitted/approved items don't show on your screen. Ask HOD or Admin to filter the audit log by your username.

5. Supervisor Manual

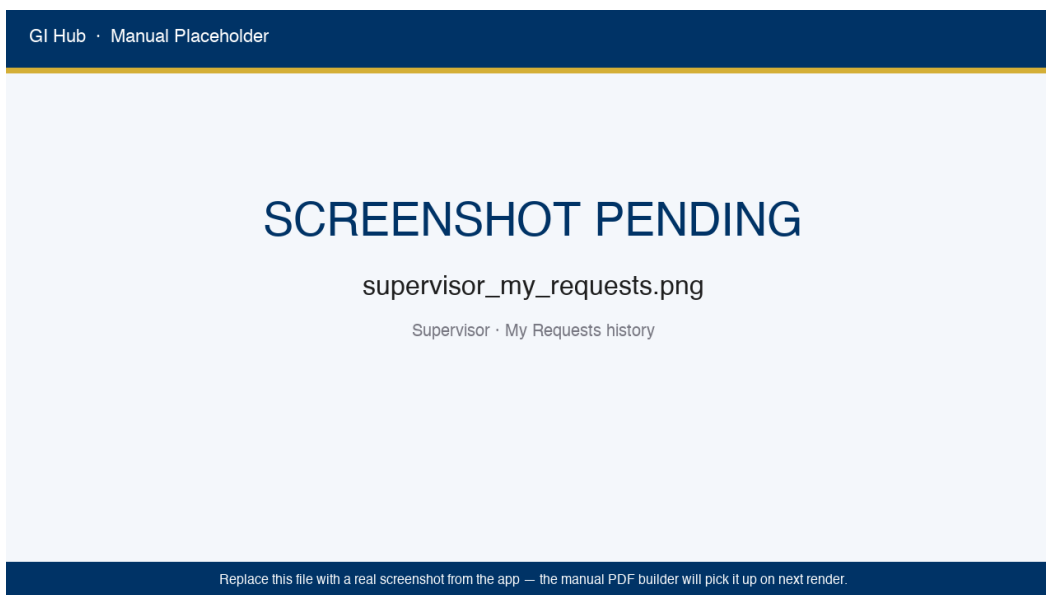
The Supervisor monitors a single site's stock, generates reports, and provides oversight. They cannot approve transactions (that's HOD).



Supervisor Portal - New Material Request form for workers

5.1 Pages visible

- Live Dashboard
- Supervisor Portal - Request Material for workers
- Entry Log (same interface as Store Keeper - see §4)

*Supervisor Portal - My Requests history with cancel option**Supervisor Portal - Intent vs Actual variance report*

5.2 Live Dashboard

The Live Dashboard is the at-a-glance view of every catalogue item with current stock, value, and status.

5.2.1 Brand header

`render_brand_header("Live Warehouse Stock Dashboard")` - gold subtitle + today's date.

5.2.2 Hero metric strip (4 cards)

Card	Source	Tone logic
Catalogue items	Count of rows in inventory	Neutral

Total stock value	Sum of Current_Stock x Unit_Cost, all sites - formatted as SAR 1,234 / SAR	Neutral. Delta: "standard cost - all sites" or "set Unit_Cost in Admin -> DB
Below minimum	Count	Green=0, Amber<10, Red>=10. Delta:
Expiring / expired	Count	Green=0, Amber<10, Red>=10. Delta:

"shelf-life clear" or "review HOD

5.2.3 Ask in plain English (AI search) - expander (if AI_ENABLED)

If Ollama is running locally with the qwen2.5-coder:7b model:

Element	Purpose
Your question text input	Natural language query, e.g., "items below minimum at site
Search button	Translates to SQL via local LLM, runs read-only
Clear button	Resets the panel
Result table	Returned rows
"Show SQL the AI generated" expander	Shows the safe SELECT the LLM produced

If Ollama is NOT running: an amber card displays setup instructions (ollama serve).

5.2.4 Burn alert banner

Appears at the top of the table area when there are items burning to zero within the configured window (default 7 days). Color-coded amber/red.

5.2.5 Main inventory grid

Column	Source	Notes
SAP_Code	inventory.SAP_Code	Primary key
Equipment_Description	inventory.Equipment_Description	Material name
UOM	inventory.UOM	Unit of measure
Total_Returned	Computed	Sum from returns table
Current_Stock	Computed identity math	Received - Consumed - Returned
Minimum_Qty	inventory.Minimum_Qty	Reorder threshold
Unit_Cost	inventory.Unit_Cost	Set via Admin -> DB Editor
Stock_Value	Computed	Current_Stock x Unit_Cost, rounded 2
Status	Computed badge	OK / Low / Below Min / Empty with colored pill via STATUS_BADGE_JS

You can sort by any column - sorting by Stock_Value shows your biggest SAR exposure first.

5.2.6 Expanders below the grid

Expander	Content
----------	---------

Stock vs Minimum Threshold	Horizontal bar chart, sorted by criticality. Each bar = item; gold line markers = minimum threshold. Color: red=empty, amber=below min, orange=close, green=ok. Capped at 20 items.
Burn Rate Forecast (30-Day)	Plotly chart with daily-burn-rate bars + vertical line at the 30-day alert threshold. Colors: red <10 days, amber <30, green >30.
Top Consumed Items	Bar chart of top 10 items by 30-day consumption, blue->gold gradient.

5.3 Reports

Supervisor sees the full Reports module but with site scope locked to their site (cannot pick "All Sites"). See §8 for the full Reports reference.

5.4 Supervisor - Use Cases

Use Case 1: Daily morning check

1. Log in -> Live Dashboard
2. Read the 4 hero cards - note any below-minimum or expiring items
3. Sort grid by Stock_Value descending -> check no over-stocked items
4. Sort by Status -> spot Empty / Below Min items
5. Open Burn Rate Forecast expander -> check the next 30 days

Use Case 2: Generate end-of-month report for management

1. Reports -> Generate Report
2. Pick Monthly Summary
3. From date: 1st of month; To: today
4. Site: locked to your site
5. Format: PDF
6. Click > Generate Report
7. Review the preview (includes SAR-value columns: Issued_Value_SAR, Received_Value_SAR, Closing_Value_SAR)
8. Click v Download PDF

Use Case 3: Investigate why an item ran out

1. Live Dashboard -> search the item (or click its row)
2. Open Burn Rate expander - see the daily rate trend
3. Reports -> Burn Rate Analysis -> date range last 30 days -> generate
4. Filter the result table for the SAP code -> see daily breakdown
5. Reports -> Daily Consumption -> narrow down which Work_Type drove the spike

5.5 Supervisor - FAQ

Q: I can only see my own site. Why can't I see other sites' stock? A: By design - site isolation. Only Admin sees all sites. If you need to know another site's stock for a transfer, ask your HOD to file a Cross-Site Request.

Q: Total stock value shows SAR 0. What's wrong? A: Items have no Unit_Cost set. Ask Admin to enter costs in Admin -> Master DB Editor -> inventory table.

Q: AI search says "Ollama not running." A: Local AI is optional. Admin must install Ollama and pull the qwen2.5-coder:7b model. Without it, the system still works - you just type into the standard grid filters.

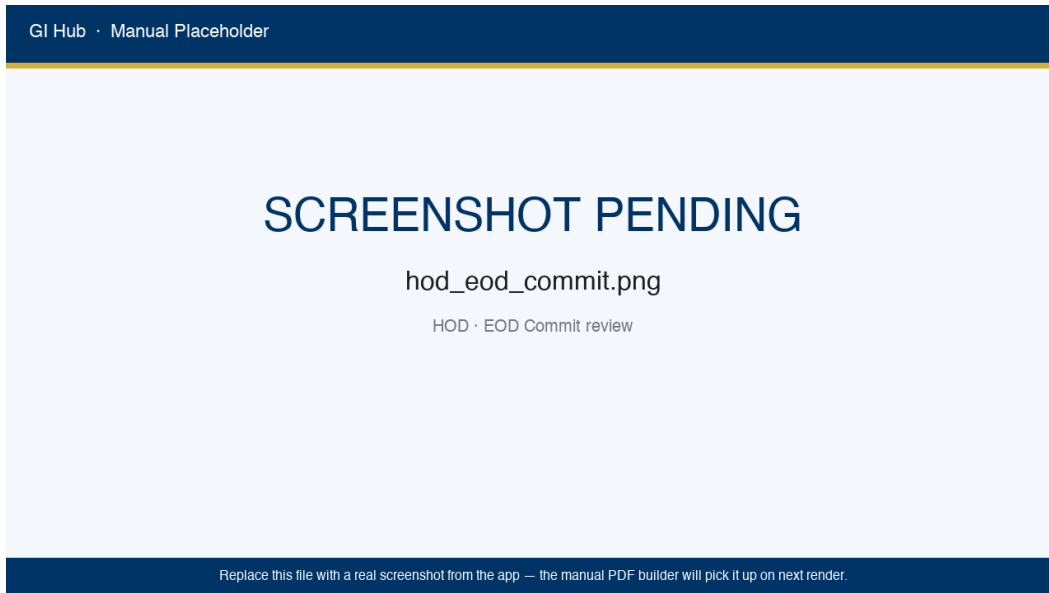
Q: The Stock_Value column is empty for some items. A: Those items have Unit_Cost = 0. The valuation is correct - they're tracked in qty but not in money.

Q: I can't approve any transactions. A: Correct. Supervisors monitor; HODs approve. Talk to your HOD.

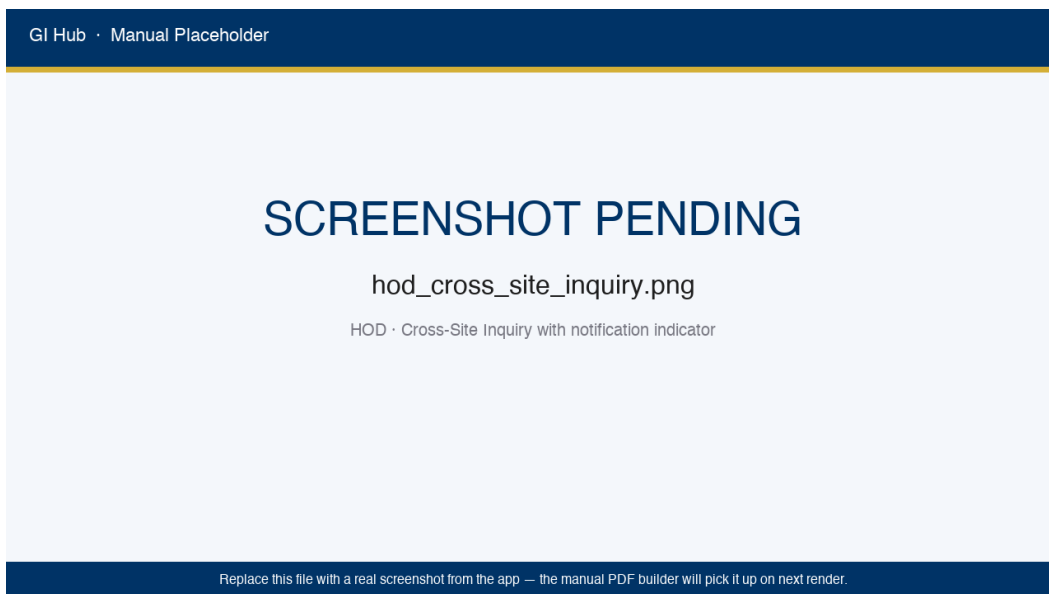
Q: Can I export the Live Dashboard grid? A: Yes - every table has a download button for CSV or Excel. For a formal, branded report use Reports and generate one.

6. HOD (Head of Department) Manual

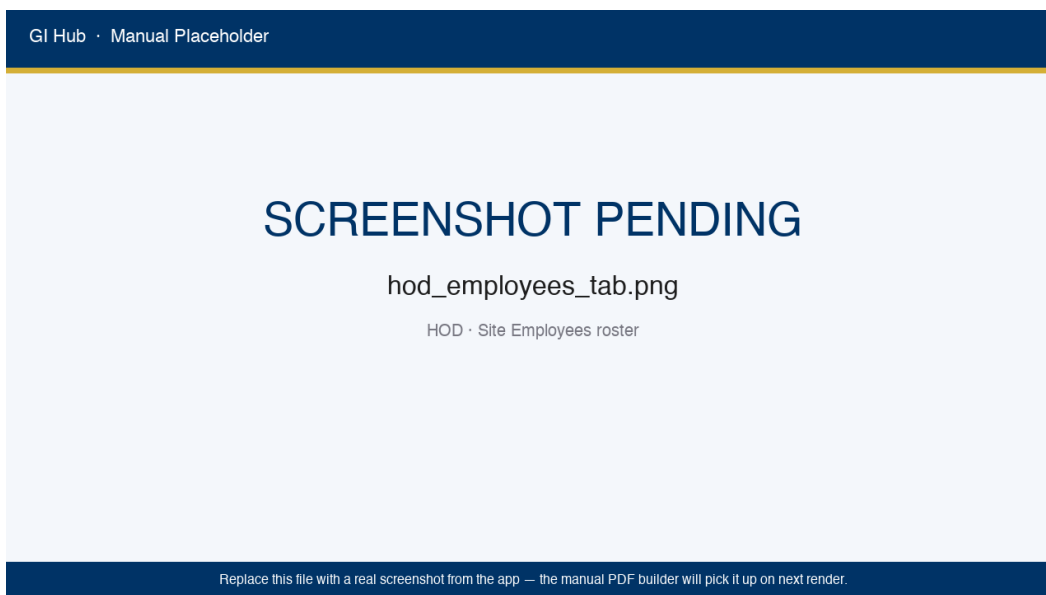
The HOD owns their site's inventory ledger. Every transaction flows through their approval. The HOD Portal has 16 tabs as of v3.0 - covering EOD, Cross-Site, Burn Rate, Receipts, Returns, Adjustments, PRs, Shelf-Life, Notifications, My Requests, Site Config, the new Employees tab (Phase 7A), DOC, QR Approval, DN Approvals, In-Transit.



HOD Portal - End-of-Day Commit review with negative-stock guard



HOD Portal - Cross-Site Inquiry with the Phase 7C "you are viewing" notification banner



HOD Portal - Site Employees roster with add/edit form (Phase 7A)

6.1 Pages visible

- Live Dashboard (see §5.2)
- Entry Log (see §4 - HODs can also stage entries)
- HOD Portal - detailed below
- Reports (see §8)

6.2 HOD Portal overview

6.2.1 Page header

render_brand_header_hod("HOD Management Portal") - purple subtitle + today's date.

- Page title: " HOD Portal" + "Managing Site: <SITE_ID>"

6.2.2 Hero metric strip (4 cards)

Card	Source	Notes
Site stock value	cached_total_inventory_value(site_id)	Delta: <SAR> consumed (30d) - site-scoped
Below minimum (site)	cached_low_stock_items(site_id) count	Green/amber/red
Expiring / expired	cached_short_dated_stock(site_id) count	Green/amber/red
Pending receipts to approve	Count from pending_receipts where status='pending_hod' at your site	Neutral or amber

6.2.3 Tab strip (10 tabs)

1. EOD Commit

2.

Cross-Site

3. Burn Rate
4. Pending Receipts
5. Adjustments
6. Purchase Requests
7. Receive Material
8. [!] Shelf-Life
9. Notifications
10. [OK] My Requests

6.3 HOD Portal -> EOD Commit

The single most consequential action in the system: committing the day's staged consumption to the permanent ledger.

6.3.1 Top section

Element	Purpose
Header	" End-of-Day Commit (<SITE>)" with intro caption

6.3.2 4-card stat strip

Card	Source
Total entries	Total pending rows at your site
Pending	Status = pending
[!] Flagged	Status = flagged
[OK] Approved	Status = approved

6.3.3 Filter pills

Pill	Action
All / Pending / Flagged / Approved / Rejected	Filters the table below to that status

6.3.4 Top action bar

Button	Action
[OK] Approve All Pending	Sets every pending row to status='approved'. Disabled when count = 0.
Commit EOD to Master	Opens the type-COMMIT modal. See §6.3.7

6.3.5 Main pending-issues table

Custom dark-themed HTML table showing every column from the staging row (joined with inventory for descriptions):

Column	Source
Date	pending_issues.Date
SAP	pending_issues.SAP_Code
Mat Code	inventory.Material_Code
Material	inventory.Equipment_Description
UOM	inventory.UOM
Qty	pending_issues.Quantity
Work Type	pending_issues.Work_Type
PR	pending_issues.PR_Number
Tank	pending_issues.Tank_No
Serial	pending_issues.Serial_No
Issued By	pending_issues.Issued_By
Issued To	pending_issues.Issued_To
Remarks	pending_issues.Remarks
Status	Colored pill (pending / flagged / approved / rejected / committed)

The table also reflects Lot_Number and FEFO_Override columns when present (visible by editing the row).

6.3.6 Row Actions panel (per-row Approve/Reject)

Below the table, top 20 actionable rows render as cards:

Element	Purpose
Card line 1	SAP - Material Code - Material name - Qty UOM
Card line 2	Date - Work Type - PR - Tank - Serial - By - To - Remarks (only fields with values)
[OK] button	Approve this single row (sets status='approved')
[X] button	Reject this single row (sets status='rejected')

6.3.7 EOD Commit confirmation modal (type-COMMIT dialog)

When you click Commit EOD to Master:

Step 1 - pre-flight: negative-stock guard

Before committing, the system checks that nothing would push stock below zero. If anything would:

Element	Purpose
Red error banner	"Cannot commit - N item(s) would go negative"
Violation table	SAP - Material - Current Stock - Trying to Consume -
Hint banner	"Common fixes: receive inbound stock first, raise a stock adjustment, or reduce the consumption to fit available stock."
Close button	Cancels - fix the staging rows and try again

Step 2 - confirm (only if no violations)

Element	Purpose
Warning banner	"You are about to commit N pending row(s)..."
Type COMMIT text input	Must type exactly COMMIT to enable the button
Cancel button	Abort
Confirm Commit button (red)	moves rows to consumption, deletes from pending_issues, busts caches, queues a post-EOD low-stock alert to the HOD if applicable.

A confirmation appears once the commit succeeds.

6.3.8 Flagged-items banner

If there are flagged rows: amber banner appears below the table reminding the HOD to verify with the store keeper before committing.

6.4 HOD Portal -> Cross-Site

For requesting material from another branch.

6.4.1 Top section

Element	Purpose
Intro caption	Explains the > 5-item escalation rule

6.4.2 Single-target inquiry flow (left/right columns)

Left column:

Element	Purpose
Select Target Branch selectbox	List of all sites except yours
Select Material selectbox	Type-ahead search across inventory
Quantity Needed number input	Min 1
Justification / Notes textarea	Why you need this transfer

Right column:

Element	Purpose
Live Stock at <target> header	Shows availability at the target site
Available Quantity metric	Live stock at target
Suggested Transfer Qty metric	min(requested, available)
Add to List button	Adds to the cart

6.4.3 Your Request Cart section

Element	Purpose
Cart table (styled HTML)	Columns: Target Site - SAP - Description - Qty - Notes

Submit All Requests to Admin button	Creates a requests row per cart item, queues WhatsApp to all admins, AND (if >5 items) escalates to the target-site
Clear List button	Empties the cart

6.4.4 Incoming Cross-Site Requests panel (bottom)

Where a HOD sees requests addressed to their site from other HODs.

Element	Purpose
Intro caption	"Requests addressed to <site>"
Incoming table (styled HTML)	From - SAP - Material - Qty - By - Notes - When
Per-row Approve / Reject controls (top 10)	[OK] Approve and [X] Reject buttons
Approval action	Calls update_request_status(status='approved') + audit
Rejection action	Same with status='rejected'

6.5 HOD Portal -> Burn Rate

Predictive analysis of which items will run out and when.

6.5.1 Compact horizontal-bar chart (top 10)

A sleek card with:

- Header row: "Monthly Consumption - Top 10" + legend chips (red=High / amber=Mid / green=Low)
- Per-row line: material name (truncated) - gradient bar (color reflects burn intensity proportional to max) - monthly figure with UOM - days-remaining badge (red if <=7d, amber if <=30d, green if >30d)

6.5.2 Burn alert banner (above the Plotly chart)

Auto-renders if items are projected to hit zero within the alert window.

6.5.3 Full Plotly burn rate chart

Daily-burn-rate bars with vertical line at 30-day alert threshold. Color-coded.

6.5.4 Detailed forecast table

Columns: SAP_Code - Equipment_Description - UOM - Current_Stock - Daily_Burn_Rate - Days_Remaining

Sortable + exportable.

6.6 HOD Portal -> Pending Receipts

Approve or reject receipts staged by Store Keepers.

6.6.1 Header + count banner

Amber banner: " N pending receipt(s) require your approval before stock levels update."

6.6.2 [OK] Approve All button

Bulk: calls `commit_pending_receipts(site_id, username)` which:

1. Iterates every pending receipt
2. Calls `process_receipt_delivery` (which creates lots if `Expiry_Date` set)
3. Posts to receipts table
4. Auto-closes PR if fulfilled
5. Deletes the staged rows
6. Busts caches, audit-logs, success toast

6.6.3 Receipt table

Columns: Date - SAP - Equipment_Description - UOM - Quantity - Supplier - PR_Number - Site_ID - Status pill

6.6.4 Per-row Reject controls (top 10)

Element	Purpose
Card line	SAP - Qty UOM - Supplier
[X] Reject button	Soft-reject (row stays as status='rejected'), preserves audit, audit-logs with reason

6.7 HOD Portal -> Adjustments

Approve physical-count reconciliations submitted by Store Keepers from the Entry Log -> Stock Count tab.

6.7.1 3-card stat strip

Card	Source
Pending	Count of pending adjustments at your site
Shortfalls	Variance < 0 count
Surpluses	Variance > 0 count

6.7.2 Pending adjustments table

Columns: # - SAP - Material - UOM - System - Counted - Variance (red for negative, green for positive) - Reason - By - When

6.7.3 Per-row Approve/Reject

Below the table:

Element	Purpose
Card line 1	SAP - Material - System X -> Counted Y - Variance
Card line 2	Reason label - submitted by - notes
[OK] button	Calls approve_stock_adjustment(adj_id, approver). Atomically posts a synthetic ledger row - if variance < 0, inserts into consumption with Work_Type=STOCK_ADJUSTMENT; if > 0, inserts into receipts with Supplier=STOCK_ADJUSTMENT. The posted_txn_ref ('C:rowid' or 'R:rowid') links back to the adjustment for full audit.
[X] button	Soft-reject. Row stays as status='rejected' for audit.

6.7.4 Recent Adjustments History (last 30)

Columns: # - SAP - Material - Variance - Reason - Status pill - Submitted By - Approved By - Ledger Ref

6.8 HOD Portal -> Purchase Requests

Manage Purchase Requests (PRs) for your site.

6.8.1 Create New PR (manual entry) - expander

Field	Purpose
PR Number	Free text, required (e.g., 3001234567)
Material selectbox	Picks from inventory; auto-fills Material_Code, Material_Name, UOM via blue-tinted info card
Requested Qty	Min 0.01
Preferred Supplier	Optional
Estimated Cost (SAR)	Optional
UOM	Auto-filled, editable
Notes	Optional
Create PR Draft button	Calls insert_manual_pr(...). Sets status='open', workflow_state='draft'

6.8.2 Upload PR PDF (auto-extract via pdfplumber) - expander

Element	Purpose
File uploader (PDF only)	Upload Purchase Request PDF
Process Upload button	extracts PR number + Material Codes (GI-XXXXXXX pattern), matches to inventory's Material_Code, inserts pr_master rows
Result banner	Green success or amber warning (if some materials unmatched)

6.8.3 Current PRs table

Columns: PR No. - SAP - Material - UOM - Qty Req - Qty Pending - Supplier - Est. SAR - Workflow pill

Pending_Qty = Requested_Qty - SUM(receipts where PR_Number matches)

6.8.4a Submit PR(s) to Logistics Portal (NEW in v3.0)

A new expander sits between the PR list and the email/PDF block. It opens the procurement-chain path - handing off your PR to the in-app Logistics queue instead of (or in addition to) the legacy email path.

Element	Purpose
Multi-select	All open PRs at your site that haven't yet been submitted to
Submit Selected to Logistics button	The <code>logistics_status</code> flips to submitted and the row appears in Logistics Portal -> Incoming PRs. A <code>pr_submitted_to_logistics</code> notification fires to the Logistics role inbox + WhatsApp (if enabled).

The legacy email path (§6.8.4) still works - your team can use it for direct-to-vendor relationships that don't go through central Logistics. Both paths coexist. The email path is marked for future deprecation once procurement chain adoption reaches the agreed threshold.

6.8.4 Notify Logistics section

When at least one PR has status='open':

Element	Purpose
Select PR for Actions selectbox	Pick from open PRs
Draft Outlook Email button	Opens email client with HTML table of pending items. Mac: opens Mail.app via AppleScript with formatted HTML table.
Download PR PDF button	Generates a PDF record with columns: Material Code - Description - Req. Qty - Received - Pending - Status. Auto-saved metadata: Site, Generated By, Timestamp.

6.9 HOD Portal -> Receive Material

For HOD-direct receipts (when a HOD logs a delivery themselves, e.g., direct purchases or HQ deliveries).

6.9.1 PR linker

Element	Purpose
Link to Open PR selectbox	Filters the material list to items in that PR, or "None (Direct Purchase)" for free-form

6.9.2 Receipt form (st.form, clears on submit)

Field	Type	Purpose
Select Material	Selectbox	From inventory or filtered to PR items
Quantity Received	Number (min 0.1)	Units arrived
Delivery Date	Date (today)	When
Expiry Date	Date (optional)	Lot expiry trigger
Logistics extras	Text inputs	All non-system columns on receipts (Vehicle_No, Driver_Name, DN_No, Supplier, Remarks, etc.)
Save Receipt button	instantly posts to receipts AND creates a lots master row if Expiry_Date is set.	

Auto-closes PR if fulfilled. Busts

6.9.3 Receipt History (last 50) caches.

Below the form, a table showing recent receipts for this site: rowid - Date - SAP - Material - Quantity - Supplier - PR_Number - Expiry_Date.

6.10 HOD Portal -> [!] Shelf-Life

Lots at risk of expiry.

6.10.1 3-card stat strip

Card	Source
Expired	Lots with days_left < 0
Critical (<=30d)	Days_left 0-30
Warning (<=90d)	Days_left 31-90

6.10.2 Action-required banner

When expired_n > 0: red banner instructing physical isolation + disposal.

6.10.3 Shelf-life table (top 30, sorted by days-left ascending)

Columns: SAP - Material - Lot - Qty - Expiry (color-coded) - Days Left - Status pill

6.10.4 Log Disposal button

Bulk-logs an audit event for the expired count. Note: this does NOT post a stock adjustment automatically - for proper inventory reduction, use the Adjustments flow with reason expired_disposal.

6.11 HOD Portal -> Notifications

Manual WhatsApp sends + alert threshold tuning.

6.11.1 Send Manual WhatsApp card

Field	Purpose
Recipient phone number	E.g., +966 5X XXX XXXX
Message textarea	Up to several hundred characters
Send WhatsApp button	Queues to whatsapp_queue with status='pending'; the background worker picks it up. Audit-logs MANUAL_WHATSAPP.

6.11.2 Alert Thresholds card

3 sliders writing to app_settings:

Slider	Range	Default	Stored key
Low stock alert (days of supply)	1-60	5	low_stock_days
Burn-rate warning (days remaining)	1-60	7	burn_alert_days
Expiry warning (days before)	1-120	30	expiry_warn_days

Save Thresholds button - persists + audit-logs UPDATE_THRESHOLDS.

> Note: The Notification Log table moved to Admin Portal -> WhatsApp Console for global visibility.

6.12 HOD Portal -> [OK] My Requests

Outbound cross-site requests YOU have made.

6.12.1 Outbound requests table (styled HTML)

Columns: # - To Site - SAP - Qty - Status pill - Created

6.12.2 Mark Incoming Transfers as Received

When at least one request has status='approved':

Element	Purpose
Select Approved Request selectbox	List of approved request IDs
Confirm Delivery Received button	Calls update_request_status(status='fulfilled') + busts caches. This is where you confirm the transfer physically arrived.

6.15 HOD Portal -> DN Approvals (NEW in v3.0)

This is your approval queue for Delivery Notes inbound to your site. The Warehouse has prepared them, Logistics has approved the delivery date - now you confirm the content (what's actually arriving, in what

qty).

6.15.1 Empty state

If nothing's pending: an empty-state card reads "No DNs awaiting your approval - They appear here after Logistics signs off the delivery date." No action needed.

6.15.2 Per-DN cards

Each pending DN renders as its own bordered container with two columns:

Left column:

- Header line: DN <number> - PO <number> - From Warehouse <id>
- Subline: DN Date - Vehicle No - Driver Name
- View lines expander: Read-only preview of every line item (Material_Code, Description, Qty, UOM, rl_bl_family, Lot_Number, Expiry_Date, Remarks)

Right column (Decide popover):

- Notes textbox (required if rejecting)
- [OK] Approve button (primary) - flips DN to pending_sk and mirrors lines into pending_receipts so SK sees them
- [X] Reject button - flips DN to rejected with your reason; Warehouse gets pinged to redo

6.15.3 When to approve

- DN qty matches what your site needs (matches the PR line you originally raised)
- RL/BL family is correct
- Lot + expiry are acceptable (not over-expiry)
- The originating PO is the right one (not a misroute)

6.15.4 When to reject

- Wrong qty (e.g. site asked for 100, DN says 200 - too much exposure)
- Wrong material (the Material_Code doesn't match the PR you raised)
- Lot has insufficient remaining shelf life
- Vehicle / driver details suggest a routing problem

Always include a clear rejection reason. The Warehouse user sees the reason verbatim in their bell inbox and on the bounced DN. Sloppy reasons cost a rebuild cycle.

6.16 HOD Portal -> In-Transit (NEW in v3.0)

A read-only window onto the procurement chain for your site. Use this tab when a user asks "when is X arriving?" or "why didn't Y arrive yesterday?". Three sub-tabs.

6.16.1 Sub-tab: Active in-transit

KPI strip at the top with counts per pipeline state:

- At Logistics - DN drafted by Warehouse, waiting on Logistics date approval
- Logistics approved - date confirmed, waiting on YOU
- Awaiting my approval - same as Logistics approved (highlighted gold)
- Pending SK receipt - you've approved, SK has it in their tab

Below the KPI strip, each in-transit DN renders as a card:

Element	Purpose
DN header	DN <number> (gold) + RL/BL chip (orange for RL, purple for BL)
Subline	PO <number> - Warehouse <id> - ETA <date> - <line count> line(s), <total qty> units
Status pill (right)	Colour-coded pipeline state
View lines expander	Read-only line preview
Request reschedule popover	Date picker (defaults to ETA + 3 days, min=today) + reason

The reschedule UI is deliberately frictionless: textarea + Submit button. Submits to Logistics.

- Date is pre-filled so a quick "+3 days" submission is one click
- min_value=today prevents accidentally picking yesterday
- Same-date submission warns instead of wasting a round-trip
- Caption " Goes to Logistics" makes the destination unambiguous

6.16.2 Sub-tab: My reschedule requests

KPI strip: Pending / Approved / Rejected counts.

Custom table showing your full reschedule history for THIS site:

Column	Source
PO No.	the linked PO
DN No.	the linked DN (if any)
From	current_date when you raised the request
Requested	the new date you asked for (gold, monospace, with arrow)
Reason	your justification (tooltip-truncated)
Status	pill - pending / approved / rejected
Decided by	Logistics user who handled it
Notes	decision notes from Logistics

6.16.3 Sub-tab: Force-closures affecting me

Read-only audit table showing every PR / PO / line that Logistics force-closed, scoped to your site. Three-way fallback join means closures on records with NULL Site_ID still appear correctly via their PR or PO parent.

Column	Source
Type	"PR closed" / "PO closed" / "Line closed" badge
Target	The closed ref (PR number / PO number / line id)
PR No.	Linked PR (if applicable)
PO No.	Linked PO (if applicable)
Reason	Logistics' force-close reason
Closed by	Logistics user who closed it
When	Timestamp

50 most recent shown.

6.13 HOD - Use Cases

Use Case 1: Approve and commit the day's consumption

1. HOD Portal -> EOD Commit
2. Review the 4-card stat strip
3. Sort/filter using the filter pills
4. For each row: read the full detail card (line 1 = essentials, line 2 = all fields)
5. Click [OK] to approve or [X] to reject individual rows
6. Optionally click [OK] Approve All Pending for a clean batch
7. Click Commit EOD to Master
8. If pre-flight blocks: read the violation table, close, fix the offending rows (reduce qty, drop, or receive stock first), then re-open
9. If pre-flight passes: type COMMIT exactly, click Confirm Commit
10. Watch the balloons.

Use Case 2: Approve a Store Keeper's physical count adjustment

1. HOD Portal -> Adjustments
2. Read the pending table - variance and reason are color-coded
3. For each adjustment, decide: did the Store Keeper count accurately?
4. [OK] Approve -> system posts a synthetic ledger row (receipt or consumption), live stock updates immediately
5. [X] Reject -> audit-logged; Store Keeper sees a status update

Use Case 3: Request material from another site

1. HOD Portal -> Cross-Site
2. Pick target site -> pick material -> see live availability
3. Add to cart, repeat
4. If cart > 5 items: an automatic WhatsApp escalation to the target HOD will fire on submit. Document why in Justification.
5. Click Submit All Requests to Admin

6.

Watch the [OK] My Requests tab - when status flips to "approved", you can mark it as fulfilled once received

Use Case 4: Approve an incoming bulk request from another HOD

1. HOD Portal -> Cross-Site -> scroll to Incoming Cross-Site Requests
2. Review the request table
3. [OK] Approve -> physical handoff begins
4. [X] Reject -> use when the request is unjustified or your stock is too tight

Use Case 5: Raise a Purchase Request

Option A: manual entry

1. HOD Portal -> Purchase Requests
2. Create New PR
3. Type PR Number, pick Material (auto-fills SAP, MatCode, UOM), Qty, Supplier, Est. Cost SAR, Notes
4. Click Create PR Draft

Option B: PDF upload

1. Upload PR PDF
2. Drop the PDF in
3. Click Process Upload
4. System auto-extracts PR number + items, creates pr_master rows

Use Case 6: Notify logistics about a pending PR

1. HOD Portal -> Purchase Requests
2. Select open PR in the email dropdown
3. Draft Outlook Email -> Mail/Outlook opens with HTML table pre-filled
4. Review/send
5. Also: Download PR PDF for your records

6.14 HOD - FAQ

Q: The EOD commit modal won't let me commit - keeps showing the violation table. A: Pre-flight is blocking because consuming the staged amount would create negative stock. Either: (a) reduce qty on the violating rows in the EOD table, (b) commit pending receipts first to top up stock, (c) raise a stock adjustment to correct any system-vs-physical discrepancy.

Q: A Store Keeper claims they submitted but I don't see anything. A: Check the filter pills (you may be filtering by Approved/Rejected only). Switch to "All" or "Pending".

Q: I approved an adjustment but the live stock didn't change. A: It should change within 30 seconds (cache TTL). Refresh the page. If it still doesn't, check audit log for the APPROVE_ADJUSTMENT entry to confirm the synthetic ledger row was posted.

Q: I got a WhatsApp about a "bulk cross-site request" - what is it? A: Another HOD at a different site has

requested more than 5 items from your site. Go to Cross-Site -> Incoming to review.

Q: My burn rate forecast looks wrong - items aren't appearing. A: Items with fewer than ~3 consumption events in the last 30 days have insufficient data and may be omitted. Increase the window via Settings if needed.

Q: I can't see a receipt I just approved. Did it work? A: Approving via "Approve All" commits all pending receipts at once. Check the Live Dashboard for updated stock. The receipt now lives in receipts table, not pending_receipts.

Q: The Outlook email button does nothing on my Mac. A: Make sure Mail.app or Outlook is installed and the system has permission to open it. The fallback is mailto: (plain-text monospace table) which works in any browser default mail handler.

Q: PR PDF upload didn't recognize some items. A: The extractor matches material codes in the format GI-XXXXXXX. Items not in your inventory's Material_Code field will be flagged in the warning. Add the missing items to inventory first.

Q: Cross-site request stuck in 'pending' - why isn't Admin approving? A: Admin sees all pending in Admin Portal -> Pending Requests. If urgent, queue a WhatsApp from Notifications. Or for >5 items, the target-site HOD can also approve directly.

Q: I want to see all FEFO overrides logged by my Store Keepers. A: Filter Admin -> Audit Logs by action FEFO_OVERRIDE. Or query the consumption table for FEFO_Override IS NOT NULL.

7. Admin Manual

The Admin is the system owner. The Admin Portal has 11 tabs as of v3.0 - the original 10 (Overview, Pending Requests, Global Sites, Users, Master DB Editor, Audit Logs, WhatsApp Console, Settings, Access Control, Reports & Bugs) plus the new Logistics Oversight tab. Admins do NOT see HOD Portal (intentional - Admin uses Admin Portal for cross-site work). Admins CAN see the Logistics Portal and Warehouse Portal as shadow access - when entering the Warehouse Portal, a sidebar dropdown lets the admin pick which warehouse to view as.

7.1 Pages visible

- Live Dashboard (global - sees all sites combined)
- Entry Log (can stage entries themselves)
- Admin Portal - detailed below
- Reports (with "All Sites" filter unlocked)

7.2 Admin Portal overview

7.2.1 Page header

render_brand_header_admin(...) - gold subtitle + pulse status chip (green "All systems operational" or amber "Degraded - see Overview" if global low-stock count > 20).

7.2.2 Hero strip (3 cards)

Card	Source
Sites managed	Count of distinct sites
Pending cross-site requests	Count from requests where status='pending'
Critical items (all sites)	low + expiry count summed globally

7.2.3 Tab strip (9 tabs)

1. Overview
2. Pending Requests
3. Global Sites
4. Users
5. Master DB Editor
6. Audit Logs
7. WhatsApp Console
8. Settings
9. Access Control

7.3 Admin Portal -> Overview

System health at a glance.

7.3.1 4-card KPI strip (technical)

Card	Source
DB size	File size of gi_database.db on disk (MB)
Users	Count of non-suspended users
Total transactions	consumption + receipts row count
Audit events	All-time count from system_audit_log

7.3.2 4-card valuation strip (financial)

Card	Source
Total stock value	all sites
Biggest-value site	cached_value_by_site().iloc[0] + share of total
30-day consumption value	cached_consumption_value(days=30)
Pending receipts value	Placeholder showing source: pr_master.Est_Cost_SAR

7.3.3 Service Health card

Per-service row with a pulse dot + status + note:

Service	Up indicator	Source
Database	Reachable, with its current size	
WhatsApp Queue	If whatsapp_queue reachable	Pending count or "queue clear"
Ollama / AI	OLLAMA_AVAILABLE if	"ready" / "not reachable"
Mail / SMTP	MAIL_ENABLED	"Outlook + mailto fallback"

7.3.4 Database Stats card

8 row counts:

- Inventory items
- Consumption rows
- Receipt rows
- Pending issues
- Pending receipts
- Open PR lines
- WhatsApp queue size
- Audit events

7.3.5 Live Activity Feed (last 12)

Per-row card from system_audit_log:

- A red, amber or green severity icon, set automatically from the kind of action recorded

—

Timestamp (gold monospace)

- Action name
- Target table pill
- "<user> - <details>" line

7.4 Admin Portal -> Pending Requests

Approve or reject cross-site material requests from HODs.

7.4.1 Pending requests data editor

Editable table with a Select checkbox column (all other columns disabled). Source: `get_pending_requests(status='pending')`.

7.4.2 Admin Notes text input

Optional when approving; required when rejecting - the requester is shown the reason, so a rejection is never unexplained.

7.4.3 Action buttons

Button	Action
[OK] Approve Selected	For each selected row: looks up Material Code/Description, updates requests.status='approved'. Then for each unique requester: queues a WhatsApp with item list + admin notes. For each unique target site: queues a "TRANSFER
[X] Reject Selected	ORDER" WhatsApp to the target HOD. Rejected Notes are MANDATORY for rejection. WhatsApp queued to each rejected requester with reason.

7.5 Admin Portal -> Global Sites

Cross-site inventory viewer (read-only).

Element	Purpose
Site selectbox	"All Sites (Global)" or pick one
Inventory dataframe	Calls <code>cached_live_inventory(site_id=...)</code>

Useful before approving a cross-site request to confirm the target site really has the stock.

7.6 Admin Portal -> Users

Delegates entirely to the system which provides:

Section	Purpose
Pending registration requests	Approve / reject self-registered users
Active users table	All users with role, site, phone
Edit user form	Change role, site, phone, suspend/activate
Add user form	Create user directly (Admin shortcut)
Password reset action	Generates temporary password, audit-logged

7.7 Admin Portal -> Master DB Editor

This is the most powerful tab. Use with care. Lets Admin view, edit, and modify any table's data and schema.

7.7.1 Table selector

Element	Purpose
Select Table dropdown	Lists every data table in the system, including lots, stock corrections, reported issues and settings.

7.7.2 Action radio

3 modes:

Mode A: View / Edit Data

Element	Purpose
Row count caption	"<N> rows in <table>"
Export as PDF button	Produces a PDF of the whole table
Data editor table	Edit any cell. Password hashes are masked as *****.
Print Label checkbox column	Only on inventory table - select items to print QR labels
Save Table Updates button	DELETE the table, INSERT_ALL from the edited dataframe. Audit-logs DB_EDIT. Busts inventory + settings
Generate QR Labels for Selected button	On the inventory - A print sheet for items where label checkbox is checked
Download QR Labels PDF button	Appears after generation

> [!] The Save action is destructive. It deletes all rows and re-inserts. If anything fails mid-write you may lose data. Industry best-practice is to make ledger tables immutable; this tab bypasses that - Admin discretion required.

Mode B: Add New Entry

Form generation depends on the table:

- For users: warns to use User Management tab instead (safer)
- For receipts: opens the full Logistics Receipt form (Site, Open PR linker, Material, Qty, Date, Expiry,

Supplier, Remarks). Posts via process_receipt_delivery -> also creates lot if Expiry set.

- For any "transaction table" (has SAP_Code, isn't inventory): 2-section form - Section 1 picks material (shows MatCode + UOM info card), Section 2 dynamically generates inputs for every editable column.
- For inventory itself: dynamic 3-column form for all editable columns.

Mode C: Manage Columns

Element	Purpose
Add Column section	Column name text input + Add button -> ALTER TABLE ADD COLUMN <name> TEXT
Rename Column section	Column dropdown + new name + Rename button -> ALTER TABLE RENAME COLUMN
Drop Column section	Pick a column and delete it. This destroys the data in that column and cannot be undone - take a backup first, every time.

> The users table is protected - schema changes are blocked here ("managed by auth.py").

7.8 Admin Portal -> Audit Logs

Complete forensic record.

7.8.1 Filter row

Filter	Source
User selectbox	DISTINCT usernames + "All Users"
Action selectbox	DISTINCT action_types + "All Actions"
Target selectbox	DISTINCT target_tables + "All Targets"
Limit selectbox	50 / 100 / 500 / 1000

7.8.2 Search text input

Substring match across details + username + action_type.

7.8.3 Audit table (styled HTML)

Columns: // severity - Timestamp - User - Action (color-coded by severity) - Target pill - Detail (truncated, tooltiped)

Severity heuristic (from action_type):

- Critical (): any action containing FAIL, REJECT, DELETE, PURGE, EMERG, ROLLBACK, RESET, DESTRUCTIVE
- Warning (): SUSPEND, REVOKE, ROTATE, FLAG, DOWNGRADE, WARNING
- Info (): everything else

Expired-row red tint on critical events.

7.9 Admin Portal -> WhatsApp Console

Outbound message queue + manual sends + thresholds + event mapping.

7.9.1 Queue stats strip (4 cards)

Card	Source
[OK] Sent	whatsapp_queue where status='sent'
Pending	status='pending'
Processing	status='processing'
[X] Failed	status='failed'

7.9.2 Send Manual WhatsApp card

Field	Purpose
Recipient phone	E.g., +966 5X XXX XXXX
Message textarea	The message to send
Send WhatsApp button	Queues to whatsapp_queue + audit-logs MANUAL_WHATSAPP

7.9.3 Alert Thresholds card (global)

Same 3 sliders as HOD Notifications (low_stock_days / burn_alert_days / expiry_warn_days). Persists to app_settings. Admin's value is the global default.

7.9.4 Event -> Recipient (current wiring) - read-only summary

A reference table showing what auto-triggers exist in the codebase:

Event	Recipient	Role
Issue staging submitted	Site HOD	hod
Pending receipt submitted	Site HOD	hod
EOD committed	Site HOD	hod
Cross-site request created	All admins	admin
Cross-site bulk (>5 items)	Target site HOD	hod
Cross-site request approved	Requesting HOD	hod
Cross-site request rejected	Requesting HOD	hod
Returnable item overdue	Store Keeper	store_keeper
New access request	All admins	admin
Access request approved	Requesting user	store_keeper
Post-EOD low stock alert	Site HOD	hod

7.9.5 Outbound Queue Log (last 80)

Styled HTML table from get_whatsapp_log(limit=80): Status pill - Recipient (monospace) - Message (truncated with tooltip) - Queued at - Sent at

7.10 Admin Portal -> Settings

7.10.1 Dropdown Manager - Work Types (expander, expanded by default)

Element	Purpose
Current Work Types caption	Comma-separated list from system_settings
New Work Type Name text input	Add a new option
Add to Dropdown button	INSERT into system_settings (category='Work_Type') + busts settings cache

7.10.2 Maintenance Mode card

Element	Purpose
Enable maintenance mode toggle	Persists to app_settings.maintenance_mode='1'
Status caption	"ACTIVE - Non-admin sessions will be told to come back later" or "Off"

The change is recorded in the audit history with who made it and when. Sign-in enforces it immediately - while maintenance is on, only Admin can sign in.

7.10.3 Database Backup card

Element	Purpose
Last manual backup caption	From app_settings.last_backup_at
Backup Now button	Calls shutil.copy2() of gi_database.db into backups/gi_database_YYYYMMDD_HHMMSS.db. Updates last_backup_at + audit DB_BACKUP.

7.10.4 Site Management

Element	Purpose
Sites table (styled HTML)	Site Name - Code (first 4 chars upper) - Users count - Status (always "Active")
Add New Site expander	Type name + button -> INSERT into system_settings (category='Site') + audit ADD_SITE

7.10.5 [!] Danger Zone (red bordered card)

Element	Purpose
Purge old draft pending_issues card	"Delete all pending_issues rows older than 30 days that are
Type PURGE to confirm	30 days - must equal PURGE to enable
Run Purge button	Executes the DELETE, audit-logs PURGE_DRAFTS

7.11 Admin Portal -> Access Control

7.11.1 Recent Sign-Ins (last 10)

Per-row card from system_audit_log filtered to LOGIN/LOGIN_SUCCESS/LOGIN_FAILED/LOGOUT:

- Pulse dot (green=success, red=failed)
- Username
- Action label
- Timestamp + details

7.11.2 Force Password Reset card

Field	Purpose
Target user selectbox	From users table
New password	Type input (password mask)
Confirm	Must match
Amber warning	"User must log in again immediately."
Reset Password button	Validates length >= 8 + match -> hash_password(new_pwd) -> UPDATE users SET password_hash + audit FORCE_PASSWORD_RESET

7.11.3 Security Policy

Read-only 2-column grid showing:

- Passwords are stored as a one-way scramble, never as text
- Sessions are held in memory only and are never written to disk
- RBAC hierarchy (store_keeper < supervisor < hod < admin)
- WAL mode + busy_timeout=5000ms
- Password min length (8 characters)
- Audit retention (indefinite, manual purge only)

7.11a Admin Portal -> Logistics Oversight (NEW in v3.0)

Cross-site, read-only window onto the entire procurement chain. For actions, jump to the Logistics Portal (shadow access) - this tab is observation-only by design.

7.11a.1 KPI strip

Six cards at the top:

Card	Source
OPEN PRs	Count - awaiting PO issuance
OPEN POs	Count from list_pos(open_only=True)
ACTIVE DNs	Count of DNs in pipeline states (pending_logistics, logistics_approved, pending_hod, pending_sk)
VENDOR RETURNS	Open returns from list_vendor_returns(open_only=True)

Card	Source
RESCHEDULES	Pending reschedule decisions
FORCE-CLOSURES	Lifetime audit count

7.11a.2 Filters

Element	Purpose
Site dropdown	"All sites" or pick one - narrows every sub-tab
Warehouse dropdown	"All warehouses" or pick one - narrows DN view

7.11a.3 Six sub-tabs

Sub-tab	What's shown
PRs	Every active PR in the Logistics queue, filterable by site
POs	Every open PO with vendor, dates, total, status, source (manual/PDF)
DNs	Every active DN with warehouse, site, status, family
<- Vendor Returns	Open returns awaiting vendor acknowledgement
Force-Closures	100 most recent force-closure records with reason + closed-by
Reschedules	Pending reschedule decisions Logistics hasn't acted on

7.11a.4 What it's NOT

- This tab cannot create / approve / reject anything. All mutation happens in role-specific portals.
- Admins who need to ACT (e.g. approve a reschedule because Logistics is on leave) should switch to the Logistics Portal - admin has shadow access there.

7.12 Admin - Use Cases

Use Case 1: Set initial Unit_Costs for valuation reports

1. Admin Portal -> Master DB Editor
2. Select Table: inventory
3. Action: View / Edit Data
4. In the data editor, find Unit_Cost column (rightmost)
5. Type SAR cost for each item
6. Click Save Table Updates
7. Wait ~60 sec for caches to refresh, then check Live Dashboard hero card "Total stock value"

Use Case 2: Approve a self-registered new user

1. Admin Portal -> Users
2. Pending requests section -> review the request
3. Edit role/site if needed
4. Click [OK] Approve -> user can now log in. WhatsApp alert auto-fires to the requester.

Use Case 3: Backup the database before a risky operation

1. Admin Portal -> Settings
2. Backup Now -> confirms file path: backups/gi_database_YYYYMMDD_HHMMSS.db
3. Note the timestamp in "Last manual backup" field
4. Now perform the risky operation (bulk import, schema change, etc.)
5. If something breaks: restore by copying the backup back over gi_database.db (with the app stopped)

Use Case 4: Investigate a suspicious series of FEFO overrides

1. Admin Portal -> Audit Logs
2. Action filter: FEFO_OVERRIDE
3. Search: e.g., a specific username
4. Review the detail column - see SAP, site, chosen lot vs FEFO-suggested, reason
5. If a pattern of unjustified overrides emerges, talk to the Store Keeper / HOD

Use Case 5: Configure WhatsApp alert thresholds

1. Admin Portal -> WhatsApp Console
2. Alert Thresholds card
3. Adjust sliders: low_stock_days, burn_alert_days, expiry_warn_days
4. Save Thresholds -> audit-logged. HOD threshold sliders read the same values.

Use Case 6: Approve a batch of cross-site transfers

1. Admin Portal -> Pending Requests
2. Review the editor table - sort by created_at, target_site, or item
3. Tick checkboxes for the rows you want to approve as a batch
4. Type Admin Notes (e.g., "Per Tuesday meeting, expedite to Site B")
5. [OK] Approve Selected -> triggers two WhatsApp dispatches per unique requester + target HOD
6. Watch the Notification log in WhatsApp Console to confirm sends

Use Case 7: Drop a deprecated column from a custom table

1. Admin Portal -> Master DB Editor
2. Select the table
3. Action: Manage Columns
4. Drop Column section -> pick the column -> Delete Column

Use Case 8: Generate QR labels for new inventory items

1. Master DB Editor -> inventory
2. View / Edit Data mode
3. In the leftmost "Print Label" column, tick items
4. Scroll to QR Code Label Generator section below
5. Click Generate QR Labels for Selected
6. Download QR Labels PDF

7.

Print on label paper, stick to shelf bins -> store keepers can now scan them on mobile

7.13 Admin - FAQ

Q: How do I add a new site? A: Admin Portal -> Settings -> Site Management -> Add New Site. New sites are immediately available for user assignment.

Q: How do I rotate a user's password without their input? A: Admin Portal -> Access Control -> Force Password Reset. Pick user, type new password twice, click reset. They'll need to log in again with the new credentials.

Q: Audit log filter has too many users - how do I find a specific user's activity? A: Use the search text input at the bottom of the audit filter row - searches across user + action + details.

Q: WhatsApp queue is stuck - messages aren't being sent. A: The notification service runs separately from the application, so the application can be perfectly healthy while messages are not going out. Ask your Admin to check it. Messages queue up in the meantime and deliver once it is running again.

Q: My DB Editor save crashed mid-way and lost rows. A: Restore from backups/. The Save action does DELETE-then-INSERT and is not crash-safe. Always backup before bulk edits.

Q: Maintenance Mode is on but non-admins can still log in. A: The toggle is remembered, and sign-in respects it - during maintenance only Admin can get in. Confirm with your Admin if you are unsure whether it is currently on.

Q: Can I delete an audit log entry? A: No - they're meant to be immutable. If you need to free space, do a SQL purge with a date filter via DB Editor -> Manage Columns... but this is itself audited.

Q: A cross-site request is stuck pending and the requester is asking why. A: Admin Portal -> Pending Requests. Confirm it's actually in the queue. If yes, approve or reject with notes. The HOD will get a WhatsApp.

Q: I want to set per-user 2FA. A: Currently unsupported in the security model (placeholder shown in Access Control). Future enhancement.

Q: How do I know if Ollama (local AI) is up? A: Admin Portal -> Overview -> Service Health card. If "Ollama / AI" pulses green, it's reachable.

Q: What's the difference between approving a Pending Receipt vs Pending Issue? A: Receipts ADD to stock (your team RECEIVED material). Issues SUBTRACT from stock (your team CONSUMED material). Both flow through approval. Issues commit via EOD (batch); Receipts commit individually via Pending Receipts tab.

Q: Stock_Value column shows nothing on the dashboard. A: Inventory items have Unit_Cost = 0. Use DB Editor to set costs.

Q: A HOD set thresholds different from mine - whose wins? A: Both HOD and Admin write to the same app_settings keys. Last-writer-wins. Decide a policy and lock it via convention.

Q: How big can the database grow before performance degrades? A: The current deployment runs comfortably at this site's user count, and the database it uses scales well beyond it. Growth in sites or

users is a hosting decision rather than a rebuild.

8. Reports Module - Detailed Reference

Available to: Supervisor, HOD, Admin. Site scope: locked for Supervisor + HOD; "All Sites" available to Admin.

8.1 Page header + 4 tabs

render_brand_header("Reports & Analytics"). Tabs:

1. Generate Report
2. Scheduled
3. AI Insights
4. Archive

8.2 Reports -> Generate Report

8.2.1 Date + filter row

Field	Purpose
From date	Start of date window
To date	End of date window
Site filter	Locked to own site for non-Admin; "All Sites" + dropdown for Admin
Format	PDF / Excel / CSV

8.2.2 Report type selector

A grid of 9 selectable cards:

Report	What it shows	Typical use
Daily Consumption	Every material issue in the window, grouped by work type	Reconciling a day's or week's usage against site activity
Monthly Summary	Per-item opening, issued, received and closing stock, with the SAR value of each	The month-end pack for finance
[!] Low Stock Alert	Materials below their minimum, with the shortfall	Building a reorder list
Burn Rate Analysis	30-day usage trend and days of supply	Deciding what to order before it runs out
Inventory Valuation	Stock value per item, highest first, with	Answering "what is on our shelves worth?"
Shelf-Life / Expiry	Items approaching expiry, critical and warning	The weekly walk-round to pull short-dated stock

Report	What it shows	Typical use
PR Status Report	Every purchase request with its stage and estimated cost	Chasing what has not arrived
[OK] FEFO Compliance	Whether issues took the oldest lot first,	Proving shelf-life discipline to an auditor
Full Audit Report	The complete activity history for the date range	Investigations, and formal audit evidence

8.2.3 > Generate Report button

Builds the report for the type, date window and site you chose. The page shows a loading shimmer while it works.

8.2.4 Preview section (after generation)

Element	Purpose
Report title + subtitle	The report type and site, above the date range covered
v Download button	Saves the report in the format you picked - PDF, Excel or CSV
Summary cards	The headline figures for the report, filled in automatically
Bar chart	Shown for the Burn Rate and Daily Consumption reports
Preview table	The rows themselves, with status colours

8.2.5 A note about apostrophes in downloaded files

Sometimes a downloaded Excel or CSV file shows an apostrophe in front of a value you typed - for example '=A1 where you typed =A1. Nothing is wrong, and nothing has been lost.

Spreadsheet programs treat anything starting with an equals sign, a plus, a minus or an at-sign as a calculation to run, not as words to display. If that text came from something a person typed into the system, opening the file could make the spreadsheet run instructions nobody intended. The apostrophe tells the spreadsheet "this is text, show it exactly as it is".

- It only ever appears in front of text that begins with one of those symbols.
- Numbers are never affected. Quantities, prices and totals stay numbers, and every total in the file still adds up correctly.
- Ordinary words, notes and material names are untouched.
- If you need the plain text back, delete the leading apostrophe in the cell.

You will see this most often on notes and remarks fields, since those are the places people type freely.

8.2.5 Email Delivery section

Field	Purpose
Recipients text input	Comma-separated emails
Send Email button	Generates and emails the report
WhatsApp button	Sends a brief summary via WhatsApp

8.3 Reports -> Scheduled

Manage automated recurring reports.

8.3.1 + New Schedule button (expands form)

Field	Purpose
Report Type	One of the 9
Frequency	Daily 06:00 / Daily 17:00 / Weekly Mon 07:00 / Monthly 1st
Format	PDF / Excel / CSV
Recipients	Comma-separated users

8.3.2 Schedule cards

Each schedule renders as a card with:

- Report icon + label
- Frequency - recipients
- Last run timestamp
- Active/Paused pill + toggle
- > Run Now button (manual trigger)
- Delete button

8.4 Reports -> AI Insights

If AI_ENABLED and Ollama is running with llama3.1:8b:

8.4.1 Header bar

+ "AI-Powered Inventory Analysis" + BETA pill + intro text. Shows "Analysing data..." shimmer during regen.

Button	Action
Regenerate	Re-runs all 5 fixed SQL probes + LLM commentary

8.4.2 5 Insight cards (collapsible)

Each card has a left-border in severity color, then:

Element	Purpose
Icon + Title	"Abnormal Consumption Spike - MAT-XXXX"
Severity pill (Critical/Warning/Positive)	Color-coded
Confidence % with progress bar	LLM-reported confidence
Right-side metric callout	Headline number with sub-label

Body paragraph	LLM-generated explanation grounded in the SQL probe result
Recommendations list	Up to 3 numbered actions
Share button	Shares with HOD team
[OK] Add to Actions button	Logs as a follow-up

The 5 fixed probes:

1. Abnormal consumption spike (vs trailing average)
2. Items approaching reorder
3. FEFO compliance rate
4. Procurement cost optimization (supplier consolidation)
5. Inventory health score

If Ollama is not running: a graceful fallback explains setup steps.

8.5 Reports -> Archive

Permanent record of previously generated reports.

8.5.1 Search row

Element	Purpose
Search archive text input	Substring match on name / type
Total counter caption	"N reports - X KB total"

8.5.2 Archive table

Columns: Report Name - Type pill (colored by report type) - Generated date - By - Format icon + label - Size - Actions

Per-row Actions:

- v Download
- Re-email
- Delete (audit-logged)

9. Automated Notifications - WhatsApp & Email & In-app Bell

The system automatically queues messages on key events. There are now THREE notification surfaces:

1. WhatsApp - messages queue and are sent by a background service, so a WhatsApp outage delays them but never loses them. There is a master on/off switch plus a per-event switch, so you can silence WhatsApp for one kind of event while leaving the in-app bell untouched for it.
2. In-app notifications bell (NEW in v3.0) - app_notifications table, surfaced via the sidebar bell described in §3.6. ALWAYS fires regardless of WhatsApp toggle.
3. Email - same Outlook / Mail.app / mailto: + SMTP paths as before.

9.1 WhatsApp triggers (auto-fire)

Event	Recipient	Trigger location	Notes
Store Keeper submits issue	Site HOD	Entry Log Submit button	Includes item list
Store Store Keeper submits receipt batch	Site HOD	Entry Log Submit button	Includes item list
Store Keeper submits stock	Site HOD	Entry Log Stock Count Submit	Includes variance + reason
Store Store Keeper performs FEFO override	Site HOD	Entry Log Add to Grid	Includes chosen vs FEFO-suggested lot + reason
HOD commits EOD	Site HOD (self)	post-commit	Only if low-stock items
HOD cross-site request created	All Admins	Submit cart	includes includes target site + count
HOD cross-site request, > 5	Target Site HOD	Submit cart	Bulk escalation
Admin Admin approves cross-site requests (batch)	Requesting HOD	Approve Selected	Includes approved items + admin notes
Admin approves cross-site requests (batch)	Target Site HOD	Approve Selected	"TRANSFER ORDER - pack
Admin rejects cross-site requests (batch)	Requesting HOD	Reject Selected	and stop rejection reason
Returnable item overdue	Borrower + Store Keeper	Background scheduler	Time-driven, not in current scope
New self-registration request	All Admins	Registration form	Includes username + role + site
Self-registration approved	Requesting user	Admin User Mgmt	"ACCESS GRANTED - Welcome"
Manual send	Free	HOD Notifications / Admin WhatsApp Console	Audit MANUAL_WHATSAPP

Event	Recipient	Trigger location	Notes
PR submitted to Logistics (v3.0)	Logistics role	HOD PR tab -> Submit PR(s) to Logistics	Event key pr_submitted_to_logistics
PO issued (v3.0)	Site HOD	Logistics -> Save PO	Event key po_issued
PO assigned to Warehouse	Warehouse users at that WH	Logistics -> Assign	Event key po_assigned_to_warehouse
Warehouse acknowledged (v3.0)	Logistics	Warehouse -> [OK] Acknowledge	Event key warehouse_acknowledged (off by default)
Warehouse received goods	Logistics	Warehouse -> Record receipt	Event key warehouse_received
Logistics approved (v3.0)	Site HOD	Logistics -> [OK] Approve DN	Event key dn_logistics_approved
DN HOD approved -> SK (v3.0)	Site SK	HOD -> DN Approvals -> [OK] Approve	Event key dn_auto_generated (reuses dn_received_by_sk)
DN received at site (v3.0)	Logistics + Warehouse	SK -> [OK] Mark as Received	Event key dn_received_by_sk (off by default)
Reschedule requested (v3.0)	Logistics	Warehouse / HOD -> Request reschedule	Event key reschedule_requested
Reschedule decided (v3.0)	Requester	Logistics -> [OK] Approve / [X] Reject	Event key reschedule_decided
Vendor return raised (v3.0)	Logistics	Any role -> <- Raise return	Event key vendor_return_raised
PR force-closed (v3.0)	Admin + originating Site HOD	Logistics -> Force-Close	Event key pr_force_closed, severity critical
PO force-closed (v3.0)	Admin + originating Site HOD	Logistics -> Force-Close	Event key po_force_closed, severity critical
Delivery reminder T-2 / T-1 / T-0 (v3.0)	Logistics + HOD + Warehouse (per DN)	Daily sweep job - see §9.3	Event key delivery_reminder_t_minus_2_minus_1_zero

9.2 Email triggers

Event	Trigger location	Mechanism
Logistics PR follow-up	HOD PR tab -> Draft Outlook Email	Mac: AppleScript-driven Mail.app with HTML table. Windows: COM Outlook with HTMLBody. Fallback: mailto:
EOD report email	mailer.py SMTP	Configurable via .env (SMTP_SERVER, SMTP_USER, SMTP_PASS)
Report delivery (scheduled)	Reports -> Scheduled	Same SMTP pipeline

9.3 In-app notifications bell (NEW in v3.0)

See §3.6 for the UI. Backed by the `app_notifications` table; queried with role + site + warehouse scoping. Per-event severity (info / warning / success / critical) drives the colour-coded left border on each card.

9.4 Delivery reminder daily sweep (NEW in v3.0)

Once a day the system looks ahead at upcoming deliveries and sends reminders two days before, one day before, and on the day itself:

Watched	When	Severity
<code>purchase_orders.Expected_Delivery</code>	T-2 / T-1 / T-0	warning / warning / critical
<code>delivery_notes.DN_Date</code>	T-2 / T-1 / T-0	warning / warning / critical

PO reminders ping: Logistics + originating Site HOD. DN reminders ping: Logistics + Site HOD + Warehouse user(s) at the receiving warehouse.

Idempotency - the sweep cannot double-fire for the same target on the same day. Two guards:

1. `UNIQUE(ref_type, ref_number, target_date, offset_days)` on the `delivery_reminders_sent` table.
2. Day-marker stored in `app_settings.delivery_reminders_last_run` - the 60-sec worker poll loop skips the sweep query entirely on second-and-later ticks of the same day.

Restarting the worker mid-day is safe. Re-running the sweep manually on the same day fires zero new notifications.

Customising the cadence - the two-day, one-day and same-day pattern is currently fixed. Making it configurable per site is on the backlog; ask if you need a different rhythm.

10. Data Model & Concept Reference

10.1 Core movement tables

Table	What it stores	Identity-math role
inventory	Master catalogue: SAP_Code (PK), Description, Material_Code, UOM, Minimum_Qty, Unit_Cost	The "items that exist" - defines what can be moved
receipts	Every received unit (post-commit). Includes Lot_Number, Expiry_Date, Supplier, PR_Number, Unit_Cost	+ adds to stock
consumption	Every consumed unit (post-EOD-commit). Includes Lot_Number, FEFO_Override, Work_Type	- subtracts from stock
returns	Tools and equipment returned to inventory	+ adds back
pending_issues	Pre-commit staging for consumption (status: draft -> pending_hod -> approved/rejected -> committed)	- does NOT affect stock
pending_receipts	Pre-commit staging for receipts	- does NOT affect stock

Identity formula: $\text{Current_Stock} = \text{Total_Received} - \text{Total_Consumed} - \text{Total_Returned} - \text{computed live in v_site_stock view.}$

10.2 Document-type tables

Table	What it stores	Lifecycle
stock_adjustments	Physical-count reconciliations	pending_hod -> approved (posts synthetic ledger row) / rejected
requests	Cross-site material transfers	pending -> approved / rejected -> fulfilled
pr_master	Purchase Request lines (manual or PDF-extracted)	status: open / closed; workflow_state: draft -> submitted -> approved -> in_progress -> received
returnable_items	Tools temporarily issued out	borrowed -> returned / overdue
lots	Lot master metadata (FEFO source-of-truth)	open -> exhausted / expired / disposed / quarantine

10.3 Supporting tables

Table	Purpose
users	Auth + role + site + phone
pending_users	Self-registration queue
system_audit_log	Immutable activity record
system_settings	Dropdown values (Work_Type, Site)
app_settings	Key/value config (thresholds, maintenance_mode, last_backup_at)
whatsapp_queue	Outbound message queue
pwa_tokens	API tokens for offline PWA
bug_reports	User-submitted issues/ideas
report_schedules	Scheduled report definitions
report_archive	Generated report metadata

10.4a Procurement chain tables (NEW in v3.0)

Table	What it stores	State machine
warehouses	Master of receiving locations	active / inactive
vendors	Supplier master	active / inactive
purchase_orders	PO header (PO_Number UNIQUE)	open -> partially_delivered -> delivered
po_items	PO line items with rl_bl_family tag	open -> partially_delivered -> delivered
po_shipment_schedule	Parsed PO Annexure rows	pending / shipped / delivered / delayed
po_assignments	Logistics -> Warehouse routing	assigned -> acknowledged -> partial / received / closed / cancelled
delivery_notes	DN header (DN_Number UNIQUE)	DN state machine: draft -> pending_logistics -> pending_hod -> pending_sk -> received (or rejected from any pending)
dn_items	DN line items	pending / received / partial / returned /
po_returns	Vendor returns (raised by any role)	open -> vendor_acknowledged / resupplied / cancelled
po_reschedule_requests	Date-change asks	pending -> approved / rejected
po_force_closures	Force-closure audit log	(terminal - write-once)
app_notifications	In-app bell inbox	read_at IS NULL (unread) -> timestamp (read)
delivery_reminders_sent	T-2/T-1/T-0 idempotency log	(terminal - UNIQUE constraint)

10.4 Views

View	Purpose
v_live_stock	Per-SAP global stock (sum across all sites)
v_site_stock	Per-(SAP, Site) live stock - the canonical "what's at this site"
v_expiring_stock	Lots/receipts in expiry buckets
v_supplier_activity	Per-supplier receipt rollups
v_lot_balance	Per-lot Received_Qty / Consumed_Qty / Remaining_Qty (identity math)

11. Status Codes, Reason Codes & Glossary

11.1 Common status pills

Status	Color	Meaning
pending	grey	Awaiting decision
flagged	amber	Has a concern (e.g., zero stock at
approved	green	Approved, ready for commit
rejected	red	Soft-rejected (row preserved for
committed	gold	Posted to permanent ledger
draft	grey	User can still edit
submitted	blue	Sent forward; awaiting next stage
in_progress	amber	Active (e.g., PR with supplier)
received	green	Fully delivered
open	blue	Active, has pending balance
closed	green	Fully fulfilled
OK / Low / Below Min / Empty	green/amber/orange/red	Stock vs minimum status
Expired / Critical / Warning	red/red/amber	Expiry buckets
sent	green	WhatsApp delivered

11.2 Adjustment reason codes

Code	Label	When to use
cycle_count	Cycle count correction	Routine periodic count
damaged	Damaged / unusable	Physical damage
expired_disposal	Expired - disposed	Expired and physically discarded
miscount_in	Miscount - found extra	Found more than system shows
miscount_out	Miscount - short	Found less than system shows
lost	Lost / unaccounted	Genuinely missing
theft	Suspected theft	Deliberate removal
return_to_supplier	<- Returned to supplier	Quality issue, sent back
other	Other (see notes)	Anything else (must explain in notes)

11.3 PR workflow states

- draft - manual creation, not yet sent
- submitted - sent to procurement
- approved - procurement greenlit
- in_progress - supplier engaged
- received - fully received

11.4 Lot statuses

- open - has remaining quantity, available for FEFO
- exhausted - fully consumed (auto)
- expired - past expiry date (manual or background)
- disposed - physically removed (manual)
- quarantine - held pending inspection (manual)

11.4a DN states (NEW v3.0)

draft -> pending_logistics -> logistics_approved -> pending_hod -> hod_approved -> pending_sk -> received With rejected as terminal from any pending state.

11.4b PO + PO line states (NEW v3.0)

PO header: open -> partially_delivered -> delivered -> closed / force_closed / cancelled PO item line: open -> partially_delivered -> delivered -> returned / closed / force_closed

11.4c Force-closure target types (NEW v3.0)

Code	Label
pr	Whole PR closed
po	Whole PO closed
po_item	Single line on a PO closed

11.4d Reschedule + vendor return states (NEW v3.0)

Reschedule: pending -> approved / rejected Vendor return: open -> vendor_acknowledged / resupplied / cancelled

11.4e RL/BL family tags (NEW v3.0)

Tag	Meaning	Detection rule
RL	Rubber Lining	Substring RL-, RUBBER LINING, RUBBER-LINING in Material_Code OR Description
BL	Brick Lining	Substring BL-, BRICK LINING, BRICK-LINING, BRICK MATERIAL
NULL	Neither family	Default

Logic in the system. NEVER a combo string - RL takes precedence if both tokens are present.

11.4f Notification severity (NEW v3.0)

Severity	Visual	When used
info (blue)	Info pings - PR submitted, PO issued, assignments	
warning (amber)	T-2/T-1 reminders, reschedule requests, rejections	
success (green)	DN approved successfully, delivery completed	
critical (red)	T-0 reminder, force-closures, urgent escalation	

11.4g Logistics-status on PR rows (NEW v3.0)

Code	Meaning
site_draft	HOD has the PR but hasn't submitted to Logistics yet
submitted	Sitting in Logistics queue waiting for PO issuance
in_po	A PO has been issued against this PR line
closed	PR fulfilled normally
force_closed	Logistics force-closed the PR with a reason

11.5 Glossary

Term	Meaning
FEFO	First-Expiry-First-Out - issue oldest-expiring lot first
EOD	End-of-Day commit (HOD action that finalizes the day's transactions)
PR	Purchase Request
DN	Delivery Note (logistics document)
OCR	Optical Character Recognition - used for bulk-staging from images
PWA	Progressive Web App - the offline mobile companion
RBAC	Role-Based Access Control
WAL	A database setting that lets many people read while one person writes. Nothing to configure - it is on
Standard cost	Per-item Unit_Cost on inventory master (vs weighted-average cost which would require receipts-history)
Identity math	Stock derived from movements, never stored as a counter

12. FAQ - Master Index by Role

12.1 General (any role)

Q: I forgot my password. A: Contact your Admin. Admin -> Access Control -> Force Password Reset.

Q: I get "Permission denied" when clicking a page. A: Your role doesn't have access. See §2.2 for the access matrix.

Q: I want to suggest a feature or report a bug. A: Any page -> sidebar -> Bug/Feature reporting dialog. Admin reviews in Admin Portal -> Reports & Bugs.

Q: The page is blank or broken. A: Hard-refresh (Cmd+Shift+R / Ctrl+F5). If it persists, check with Admin - they can see audit logs to diagnose.

12.2 Store Keeper - see §4.8

12.3 Supervisor - see §5.5

12.4 HOD - see §6.14

12.5 Admin - see §7.13

13. 2026-06 Feature Update - What Changed

This section documents the upgrades shipped in the 2026-06 release. Everything above remains accurate except where this section overrides it.

13.1 Field-level changes

Change	Where	Behaviour
All form fields mandatory	Entry Log (Consumption / Receipt Staging), HOD Receive Material, Admin Add Entry forms	Every text/number/select input is required. Validation lists missing fields
Expiry Date is optional	SK Receipt Staging, HOD Receive Material	Mandatory (Optional). Leave blank for non-perishable items.
Remarks, Tank No., Serial No., PR Number	Same forms	No longer optional.

13.2 Live Dashboard column order

The Live Dashboard table now renders in this order (when columns exist):

SAP_Code -> Material_Code -> Equipment_Description -> UOM -> Opening_Stock -> Receipt -> Consumption -> Return -> Closing_Stock -> Minimum_Qty -> Unit_Cost -> Stock_Value -> Category -> Status

- Opening_Stock is now a configurable column on inventory. Default 0; admin can edit in DB Editor.
- Identity formula updated to $\text{Closing_Stock} = \text{Opening_Stock} + \text{Total_Received} - \text{Total_Consumed} - \text{Total_Returned}$.
- Material_Code now appears after SAP_Code in every report (Daily Consumption, Daily Receipts, Monthly Summary, PR Status, etc.) and in the HOD Pending Receipts approval list.

13.3 Entry Log access - Store Keeper only

The Entry Log page is now visible only to the store_keeper role. HODs review submissions in HOD Portal; Admins use Admin Portal. The page is hidden in the sidebar for other roles.

13.4 EOD Commit - checkbox confirmation

The "Confirm EOD Commit" dialog no longer requires typing COMMIT. Tick the confirmation checkbox and click Confirm Commit. Cancel still drops all pending state.

13.5 Material Category

Every inventory item now carries a Category. Categories: Consumable, Equipments, Utilities,

Maintenance, Others (default), Rubber materials, Tools, QC items.

- Admin Portal Add New Entry -> renders Category as a selectbox.
- Reports page -> Filter by Category dropdown alongside the SAR / cost-columns toggle. "All Categories" disables the filter.

13.6 Rubber MTC workflow

When a Store Keeper stages a receipt and the selected material's category is Rubber materials, the system shows:

- MTC Number text field (e.g. MTC-2026-001234)
- MTC Document file uploader (PDF / JPEG / JPG / XLSX)

Either field can be blank - the receipt still goes through. What changes is HOD-side visibility:

- HOD Portal -> Pending Receipts: a red banner lists rubber items received without an MTC.
- Click Draft Logistics Email to open a pre-filled email to the logistics team listing SAP, description, lot, qty.
- Sending (or clicking "Mark all as sent") flips the rubber rows to sent_to_logistics.

The logistics email recipient defaults to LOGISTICS_EMAIL env var (logistics@generalindustries.net if unset). On Windows it opens Outlook; on macOS it opens Mail.app; on Linux it opens the default mailto handler.

13.7 Document attachments - Entry Log + HOD DOC tab

Store Keepers can attach reference documents (PDF / JPEG / JPG / XLSX) on:

SK form	Doc number used	Notes
Consumption Log	Auto = DDMMYY of the date	Pick scope: "Whole entry (batch)" or "Specific date".
Receipt Staging	DN No. of the row (or manual override)	Falls back to DN-DDMMYY if no DN_No found.

Files are stored as BLOBs inside the database (authoritative copy) and mirrored to uploads/<Site_ID>/<doc_type>/<doc_number>/ for local browsing. The disk mirror is gitignored; only the DB BLOB is portable.

HOD Portal -> DOC is a new tab with three sub-tabs: Consumption / Receipt / <- Return. Each shows period (From/To dates) and Doc Number text filters, with a per-file download button.

The <- Return sub-tab pulls from the new Return Items workflow (see §13.10), not from Returnable Items.

13.8 QR Label approval flow

The Admin DB Editor's QR generator (single-user, single-item) is unchanged. The new flow is two-step:

1.

Store Keeper -> Entry Log -> QR Label Request (new tab)

- Multi-select materials in one form.
- Per-item label quantity.
- Click Submit Batch for HOD Approval.

1. HOD Portal -> QR Approval (new tab)

- Pending sub-tab: select rows via checkbox, then [OK] Approve Selected or [X] Reject Selected.
- [OK] Approved sub-tab: Download QR Labels PDF for ALL approved generates one consolidated PDF.

13.9 Returnable Items - clarification

The Returnable Items tab is for temporary tool loans only (e.g. issuing a torque wrench to a worker who'll return it before EOD). It is not a way to return stock to the warehouse. No DN No., no document attachment.

For real returns (defective material going back to logistics), use the new Return Items tab - see §13.10.

13.10 Return Items workflow (NEW)

The new <- Return Items tab (between Receipt Staging and Returnable Items in the Entry Log) handles real returns to the warehouse / logistics.

Store Keeper flow

1. The material picker is restricted to materials received in the last 30 days at the user's site.
2. If multiple receipts exist for the same SAP code, the SK is asked which receipt is being returned (Date / DN No. / Received Qty).
3. The system shows a locked summary of the original receipt: Date, DN No., PR, Lot, Received Qty.
4. The SK enters:
 - Return Quantity (capped at the original Received Qty)
 - Reason (work-types dropdown)
 - Return DN No.
 - Attachment - mandatory: the Return DN + any photos
9. To return material older than 30 days, tick "Override 30-day window" - the picker widens to 12 months and an Override Justification field appears. This routes to HOD as an explicit override request.
10. Submit -> request lands in HOD Portal <- Returns tab. A WhatsApp ping goes to the site HOD if a phone number is on file.

HOD flow

1. HOD Portal -> <- Returns lists every pending return with a card per row.
2. Rows that required an override are highlighted in red and show the SK's justification.
3. [OK] Approve -> writes a row to the returns ledger (so Current_Stock reduces by the returned qty, the dashboard Return column ticks up, and the entry shows up in monthly / consumption reports). Then

automatically opens the logistics email draft with item, qty, reason, and the original receipt's DN/PR/Lot context.

4. [X] Reject -> marks the request rejected. The SK sees this in their request history.

Dashboard / report impact

- The Return column on the Live Dashboard reflects approved returns (since returns is the source of truth).
- All existing reports (Daily Consumption, Daily Receipts, Monthly Summary, Audit) include returns via the same identity math.
- The HOD DOC tab <- Return sub-tab lets the HOD browse all attached return documents.

13.11 Per-site Work Type and Tank No.

HOD Portal -> Site Config lets the site HOD add or delete Work Types and Tank Numbers scoped to their site. Empty per-site lists fall back to the global defaults.

13.12 WhatsApp worker - startup fix

The notification service now starts instantly instead of taking tens of seconds, because it loads its delivery components only when the first message actually needs sending.

If startup appears to hang for more than a few seconds after an update, the update was probably applied without its accompanying dependency step. Ask your administrator to re-run the update procedure.

13.13 Category rename - Rubber Materials -> Surface Shields

The category that triggers the MTC-required workflow on Receipt Staging is now Surface Shields instead of "Rubber Materials". Everywhere the system used to look for Rubber materials, it now looks for Surface Shields. The behaviour is identical:

- SK selects a Surface Shields item on Receipt Staging -> MTC Number + MTC File uploader appear.
- Missing MTC = HOD sees the red banner on Pending Receipts with Draft Logistics Email.

Which category requires a material certificate is a single setting, so extending the requirement to another category is a one-line change for your administrator rather than a development task.

13.14 WBS Master + WBS-aware Entry Log + WBS Report

- HOD Portal -> Site Config -> WBS Numbers lets the HOD add, close, or re-open WBS numbers for their site. Each WBS carries an optional Description and an active / closed status.
- Entry Log -> Consumption Log and Receipt Staging show a WBS Number dropdown filtered to the SK's site. If the HOD hasn't added any WBS yet, the SK sees a warning and a free-text fallback so work isn't blocked.
- Reports -> WBS Report rolls everything up by WBS for a chosen date range, scoped to the user's site. Columns: WBS_Number, Consumption_Rows, Consumption_Qty, Consumption_Value_SAR,

Receipt_Rows, Receipt_Qty, Receipt_Value_SAR. Sorted by consumption value descending.

- DB column wbs is now self-healed on consumption, receipts, pending_issues, pending_receipts. Old rows show as (no WBS) in the report; new rows carry the picked WBS through SK -> HOD commit untouched.

13.15 Site_ID badge in sidebar

Every signed-in user now sees their site code as a gold pill in the sidebar user card, next to the role label. This makes "wait, which site am I logged into?" impossible to get wrong, which matters when an Admin shadows a site for support.

13.16 Live Dashboard - live-typing filter on key columns

The dashboard filter row no longer waits for Enter. Typing into any of the four searchable columns - SAP_Code, Material_Code, Equipment_Description, Category - narrows the table on every keystroke (~180 ms debounce). The numeric columns deliberately have no filter input: text filters on numbers don't help and they made the page laggy.

Header reads "Filters (live) - searchable on SAP / Mat Code / Description / Category".

The package streamlit-keyup powers the live updates; if it's not installed, the dashboard falls back to plain st.text_input (still works, just needs Enter).

13.17 Sidebar Hub Assistant - role-aware Q&A

A new sidebar expander Ask Hub Assistant lets any signed-in user ask plain-English questions about the system. The answer is generated by the local Ollama llama3.1:8b model. Role filtering happens at the RAG context layer, not just the system prompt - a Store Keeper's question never sees the Admin chapter, so the model physically cannot answer about admin features even if asked.

Section visibility:

- Store Keeper: §1, §2, §3, §4, §11, §13
- Supervisor: SK list + §5, §8
- HOD: Supervisor list + §6, §9
- Admin: everything

The widget streams the answer token-by-token. Click Clear to reset.

13.18 GMT+3 (Asia/Riyadh) timestamps

Timestamps shown in the UI are now Riyadh local time. Background:

- New records are stamped in Riyadh time directly, because the server itself is set to that time zone.
- Times stored by the database itself are converted to local time when they are displayed.
- Affected views so far: Admin Pending Cross-Site Requests, Admin Audit Logs, Admin Live Activity

Feed, Admin WhatsApp Console, HOD Cross-Site Incoming, HOD Pending Receipts. Other tabs still show UTC; tell the admin to file a ticket if any specific column matters and isn't converted yet.

A site in a different time zone is a single setting for your administrator - no code change is needed.

13.19 Self-hosting

Everything needed to run the system on your own hardware ships with it: the background services, the installer, the restart and uninstall procedures, the nightly backup, and a full setup playbook that takes about 45 minutes end to end.

Section 17 covers what this means for you day to day. The procedures themselves are in the deployment runbook, for whoever administers the server.

13.20 WhatsApp notifications

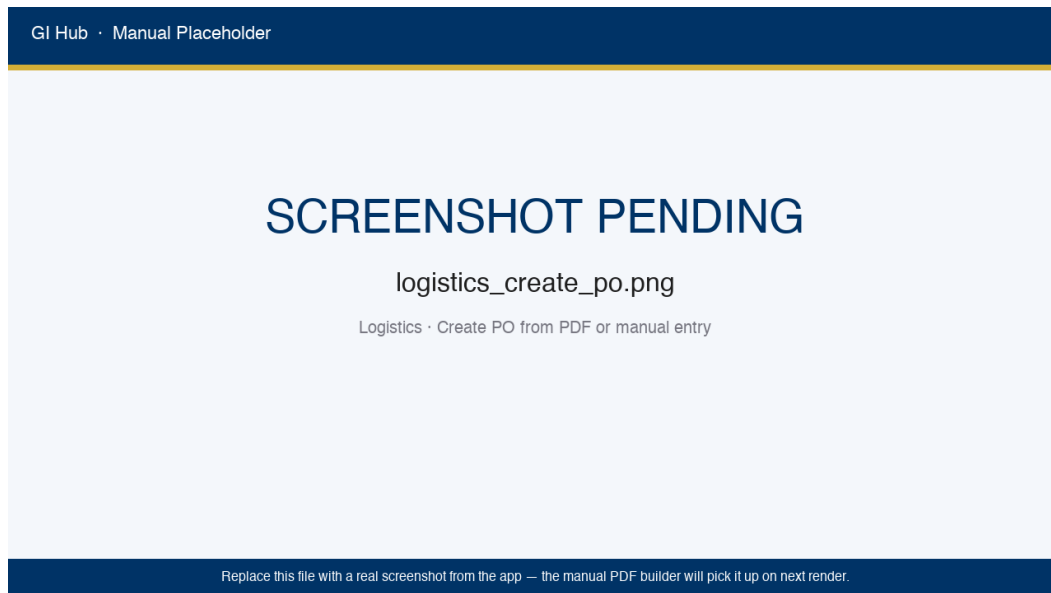
WhatsApp delivery is handled by a background service that runs independently of the application. Two things follow from that, and both are improvements:

- A WhatsApp outage never blocks work. Messages queue and send once the channel is back. Nothing is lost and nobody is held up.
- In-app notifications are unaffected. The bell in the header is filled by the application itself, so it keeps working even when WhatsApp does not.

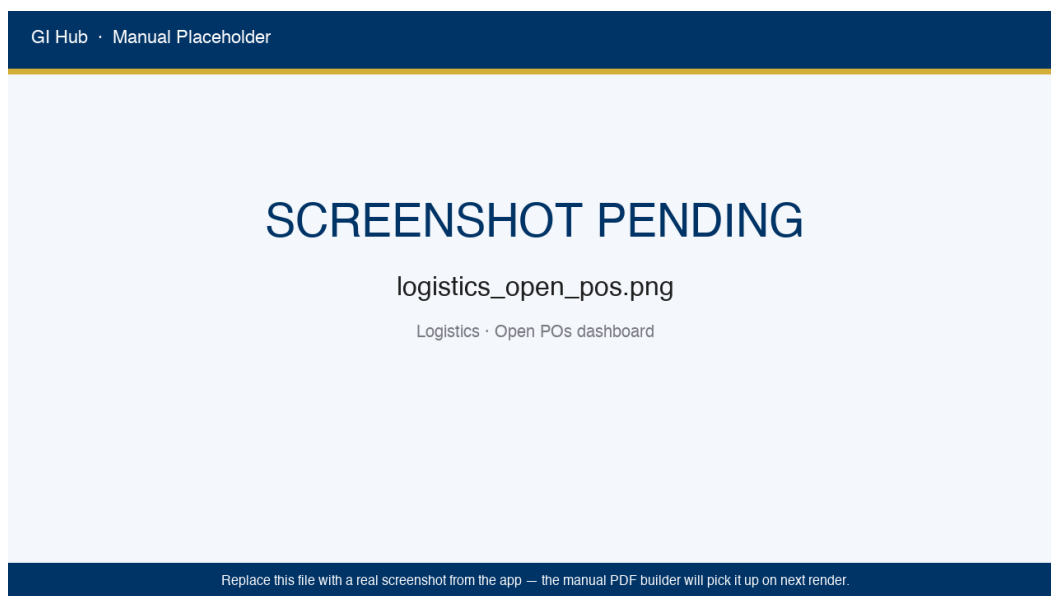
If messages stop arriving, see section 17.7.

14. Logistics Portal Manual

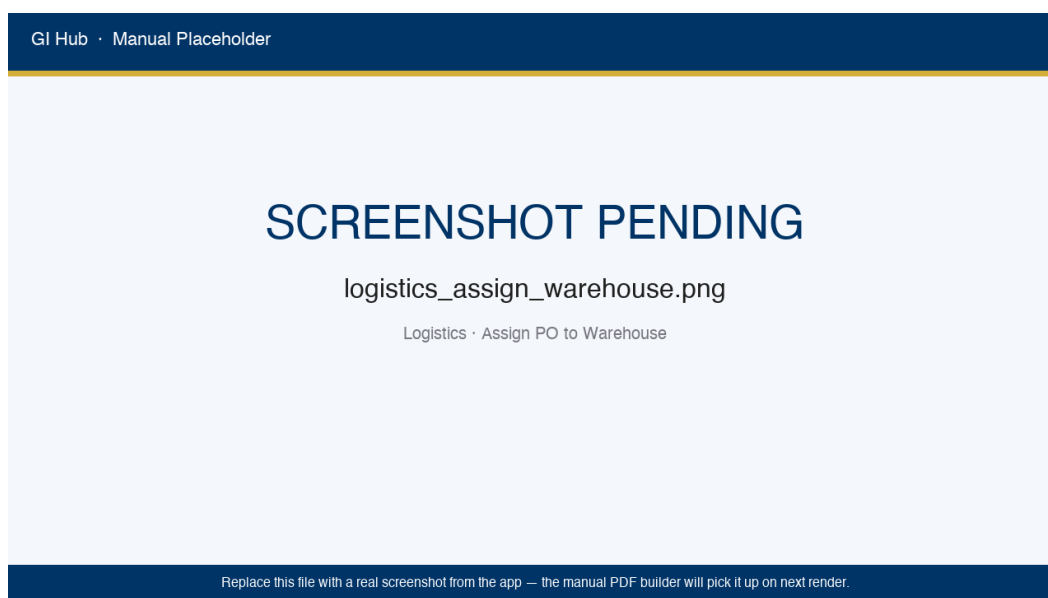
The Logistics Portal sits between Site HOD (who creates PRs) and Warehouse (which physically receives goods). Role-locked to {logistics, admin} - exact-role lock means no other role inherits access via the hierarchy. Eight tabs.



Logistics Portal - Create PO from PDF upload or manual entry



Logistics Portal - Open POs dashboard with status pills

*Logistics Portal - Assign PO to a Warehouse*

14.1 Pages visible to Logistics

- Live Dashboard (read-only - all sites)
- Logistics Portal (this page)
- Reports (incl. the 3 new procurement reports)

14.2 Tab 1: Incoming PRs

Site HODs submit PRs to Logistics from their HOD Portal -> PR tab. They land here.

14.2.1 Site filter

Dropdown of all sites + "All sites". Defaults to all.

14.2.2 Hero strip (3 cards)

Card	Source
OPEN PRs	Active queue count
TOTAL QTY	Sum across all open PR lines
SITES	Distinct sites currently submitting

14.2.3 Queue table

A table with columns: PR No. - Site - Lines - Total Qty - Submitted - Earliest Delivery - Status. Sortable, filterable, exportable.

14.2.4 Per-PR drilldown

Selectbox under the queue table. Pick a PR + Site combination and a section card appears:

Card row	Source
PR Number	the row
Site	the row
Lines	line count
Total Qty	summed across lines
Earliest Delivery	min of Delivery_Date across lines

Below: full line items grid with Material_Code, Material_Name, Requested_Qty, UOM, WBS_Number, Network, Plant, Delivery_Date, Supplier, Est_Cost_SAR.

14.2.5 Use this PR to create a PO button

Loads the PR into Tab 2 (Create PO). Selectbox state is preserved so you can switch tabs without losing the chosen PR.

14.3 Tab 2: Create PO

Two sub-tabs: Manual entry and PDF upload. Both funnel into the same purchase_orders insert path.

14.3.1 Vendor picker

Selectbox of all active vendors <code> - <name>, plus a " Add new vendor" option that opens an inline expander with Vendor Code, Name, Address, Default Inco Terms, Default Payment Terms. Save -> re-renders the parent form with the new vendor pre-selected.

14.3.2 PO header form

Three-column layout:

Column	Fields
Left	PO Number * - PO Date - PO Type
Middle	PR Number - Quotation No. - Quotation Date
Right	Expected Delivery - Inco Terms - Payment Terms

Plus a second three-column row for Contact (vendor), Contact Email, Mobile, Our Reference, Your Reference, Our Email.

Vendor defaults (Inco / Payment Terms) auto-fill from the master row but remain editable.

14.3.3 Manual sub-tab: line items

If a PR was loaded via the Tab 1 shortcut, every PR line pre-fills with Include = True and zero Unit_Price + Total_Price (you set them). Otherwise an empty editable grid opens.

Editable columns: Include - Material_Code - Description - Qty - UOM - Unit_Price - Total_Price - WBS_Number - Network - Plant. Rows with Include = False are dropped on save.

14.3.4 PDF upload sub-tab

Element	Purpose
File uploader	Drop the PO PDF
Extract from PDF button	regex-based extraction of header (PO Number, Vendor, Inco/Payment, Quotation refs, totals) + line items (Sr. No, Material_Code, Description, Qty, UOM, Unit_Price, Total_Price) + PO Annexure delivery schedule (Shipment N
Review extracted PO	Editable preview of every extracted field. The header form pre-fills. The line items grid pre-fills. Edit anything before saving.
Delivery schedule	If Annexure parsed, a table shows shipment_no - material_group - target_date
Save PO (from PDF)	Persists the PO + items + shipment schedule. The original PDF is stored as a BLOB on the purchase_orders row (attachment_blob/_name/_mime) for audit.

On PO numbers with X-masking: the sample PDF has the last 4 digits masked as XXXX for security. The extractor preserves whatever is on the page verbatim. In production, vendors send full 10-digit numbers and those pass through unchanged.

14.3.5 Side-effects on save

- New row in purchase_orders with status 'open'
- One row per item in po_items with rl_bl_family auto-tagged
- Linked PR rows (matching PR_Number + Site_ID) flip to logistics_status='in_po' so they leave your Incoming PRs queue
- Site HOD gets po_issued notification (in-app + WhatsApp gated)

14.4 Tab 3: Open POs

Browse + drill into every open PO.

14.4.1 Filters

Site dropdown - Vendor dropdown - PR Number exact-match textbox.

14.4.2 KPI strip

OPEN POs count - Total value SAR - PENDING (not yet delivered) count.

14.4.3 Per-PO drilldown

Select a PO -> section card with PR Number, Vendor, Inco/Payment Terms, PO Date, Expected, Source (manual/pdf_upload), Status, Total. Below: items grid with an RL/BL family chip per row, the parsed delivery schedule (if any), and a list of warehouse assignments.

14.5 Tab 4: Assign to Warehouse

Route a PO (or a subset of items) to a warehouse for receiving.

14.5.1 PO + Warehouse pickers

Two-column row: PO selectbox - Warehouse selectbox (shows <id> - <name>).

14.5.2 Items selector

Editable grid showing every PO line with an Include checkbox (default True). Disable lines you don't want to route - they stay with this PO for a future assignment.

14.5.3 Header

Expected Delivery date + Notes (visible to Warehouse).

14.5.4 Assign to Warehouse button

- Subset = if all items included, encoded as None (cleaner audit)
- Subset = if partial, encoded as JSON list of po_items.id in po_assignments.items_subset_json
- Warehouse user(s) at the chosen WH get a po_assigned_to_warehouse notification (in-app + gated WhatsApp)
- PO header's Expected_Delivery auto-fills with this date if not set
- Audit log entry: ASSIGN_PO_TO_WAREHOUSE

14.6 Tab 5: Reschedules

Incoming reschedule requests from Warehouse / Site HOD.

14.6.1 Empty state

If no pending requests: empty-state card.

14.6.2 Per-request card

Element	Purpose
Header line	PO <number> - DN <number> - From <current_date> -> <requested_date>
Subline	requested_by + role + reason
Decide popover	Decision notes textbox + [OK] Approve / [X] Reject buttons

Approval auto-pushes the new date to: ^(reject requires reason) the PO header Expected_Delivery, the po_assignments.Expected_Delivery, and the DN DN_Date (if linked).

14.7 Tab 6: Force-Close

Use sparingly. Force-closing notifies Admin + originating Site HOD immediately with the reason.

14.7.1 Target radio

PR (entire) - PO (entire) - PO line (single item)

14.7.2 Reason textbox

Mandatory, minimum 3 characters. Logged verbatim to po_force_closures + audit.

14.7.3 Target picker (changes per radio)

- PR: dropdown of open PR numbers in Logistics queue
- PO: dropdown of open POs
- Line: PO dropdown -> then line dropdown showing only open / partially_delivered lines

14.7.4 Force-close button

Single confirm-and-execute. Behind it:

- pr_master.status='closed' + logistics_status='force_closed' (for PR target)
- purchase_orders.status='force_closed' + line statuses for force_closed (for PO target)
- po_items.line_status='force_closed' (for line target)
- New row in po_force_closures with reason
- Notifications fan-out: Admin (recipient_role='admin') + originating Site HOD (recipient_role='hod', recipient_site=<site>), severity = critical

14.7.5 Recent force-closures (audit)

A table of the last 50 closures. Read-only.

14.8 Tab 7: <- Vendor Returns

Raise a return to the vendor against a PO. Returning a line REOPENS the PO so it shows in your active queue again.

14.8.1 PO picker + scope

PO selectbox (all POs incl. closed). Then radio: Whole PO or Single line. If single line, a line picker appears.

14.8.2 Return details form

- Return quantity (number, > 0)
- Reason (textarea, mandatory)
- Expected resupply (date picker, defaults to today + 14)
- Notes (optional)

14.8.3 <- Raise vendor return button

- New row in po_returns with raised_by_role='logistics'

po_items.Returned_Qty bumps; line_status flips back to partially_delivered or open

- If the PO had been closed, header flips back to partially_delivered and closed_by / closed_at / close_reason are nulled
- Notification: vendor_return_raised to logistics inbox + WhatsApp gated

14.8.4 Open returns table

A table of all open returns. Read-only.

14.9 Tab 8: History

Read-only archive. Two sub-tabs: Closed POs and Closed PRs.

- Closed POs: status in closed / force_closed / cancelled
- Closed PRs: any PR where pr_status='closed' OR logistics_status in force_closed/closed/in_po

For force-closures with the reason history, use Tab 6 -> Recent force-closures table.

14.10 Logistics - Use Cases

Use Case 1: Issue PO from a fresh PR

1. Incoming PRs -> pick PR -> Use this PR to create a PO
2. Create PO -> Manual entry -> pick / add vendor, fill header, edit Unit Price per line
3. Save PO -> balloon animation -> notification fires to Site HOD

Use Case 2: Issue PO from a vendor's emailed PDF

1. Receive PO PDF from vendor (sample format: Create PO -> PDF upload)
2. Drop the PDF -> Extract from PDF
3. Review extracted header + line items + delivery schedule
4. Correct anything the parser missed
5. Save PO (from PDF) - original PDF is archived

Use Case 3: Route a PO to a warehouse

1. Assign to Warehouse -> pick PO + warehouse
2. Disable lines you want to route separately later
3. Pick Expected Delivery, add a routing note
4. Assign - warehouse user is pinged

Use Case 4: Handle a Warehouse reschedule ask

1. Reschedules -> review the request (reason text + current vs requested date)
2. Decide -> notes + [OK] Approve
3. PO Expected_Delivery auto-updates; Warehouse + requester get the decision notification

Use Case 5: Force-close a stale PR

1. Force-Close -> radio = PR (entire)
2. Write the reason (e.g. "Project cancelled by site management 2026-06-15")
3. Pick PR -> Force-close PR
4. Admin + Site HOD see a critical notification in their bell

14.11 Logistics - FAQ

Q: A PR appears in Incoming PRs but I can't issue a PO against it - Material_Code is wrong. A: PR lines come from the Site HOD's inventory catalogue. If the Material_Code is wrong, the SK created it wrong upstream. Don't fix in your PO - bounce the PR back via WhatsApp / chat, ask the site to re-submit. There's no in-app "reject PR" button (yet) because PR rejection should be a conversation, not a click.

Q: PDF extraction picked up wrong qty. A: The Review preview is editable. Always check Qty / Unit_Price / Total_Price before saving. The extractor is calibrated against the General Industries sample layout; other vendor templates may need template additions.

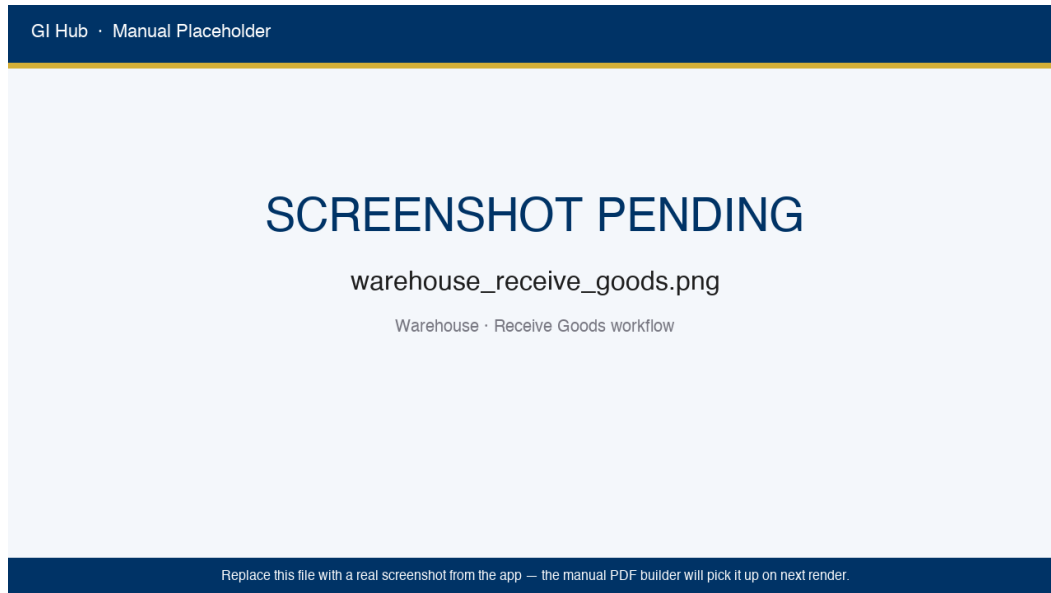
Q: I want to assign a PO to multiple warehouses. A: Two separate Assign actions - one per warehouse. The items_subset_json field on each assignment row keeps them distinct.

Q: Why can't I see prices in Tab 4 / on assignment cards? A: You CAN see prices in Tab 3 (Open POs drilldown). Tab 4 (Assign to Warehouse) shares the read with Warehouse users, so prices are hidden there for consistency.

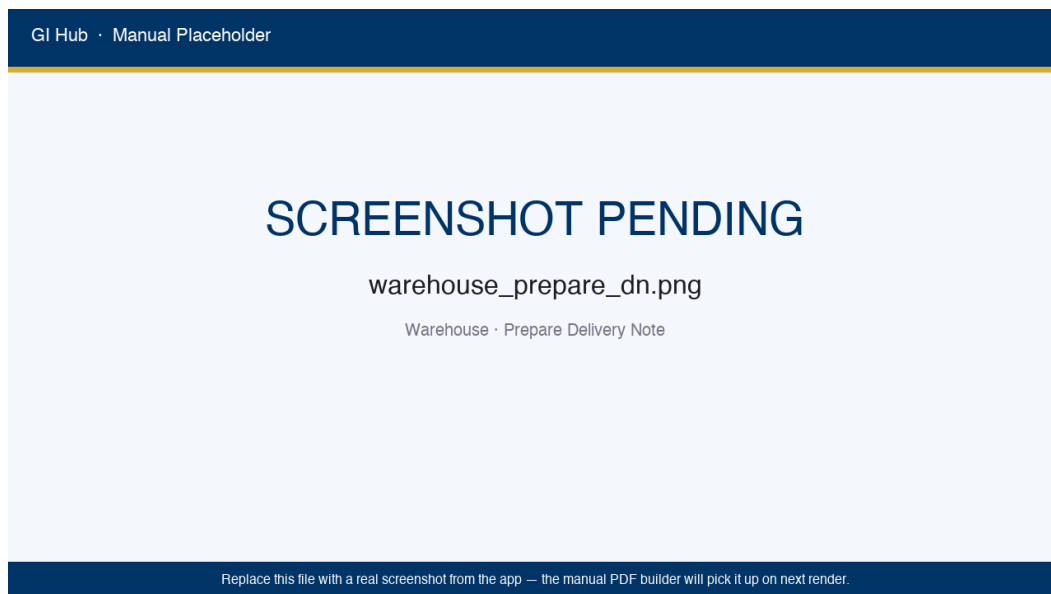
Q: Force-closure undo? A: Not yet. Once force-closed, you'd have to either raise a Vendor Return (reopens the line) or admin-edit via Master DB Editor. A 24-hour undo window is on the v3.0 backlog.

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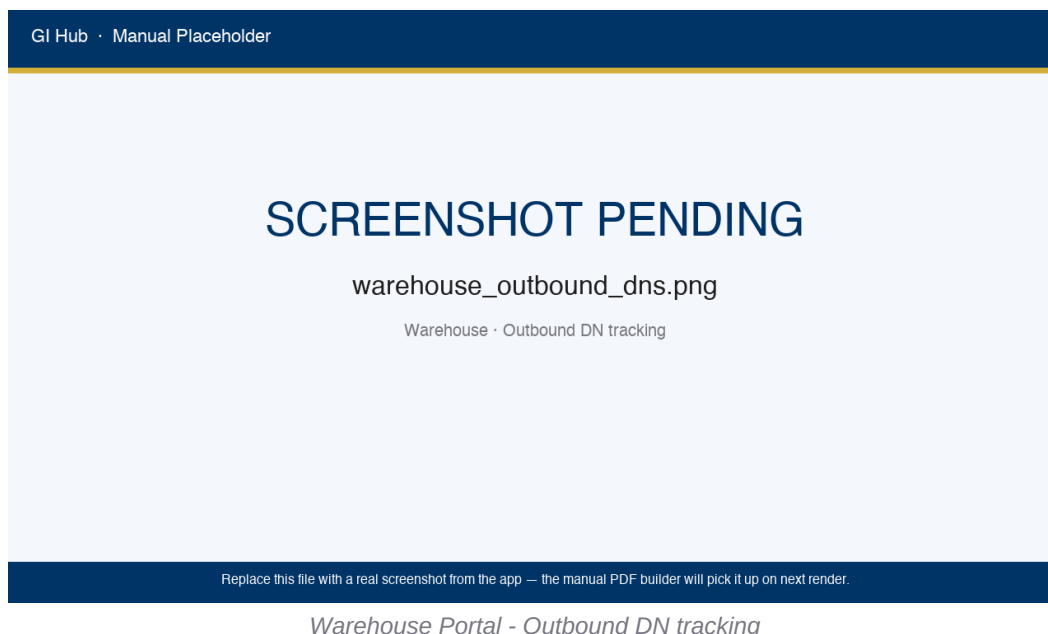
The Warehouse Portal is the physical-receiving and DN-preparation side. Role-locked to {warehouse_user, admin}. Six tabs. Prices are completely hidden in every view - three independent enforcement layers guarantee Unit_Price, Total_Price, Total_Amount, Freight_Charges, Handling_Charges, Discount_Amount, Amount_In_Words are never visible to a warehouse user.



Warehouse Portal - Receive Goods workflow



Warehouse Portal - Prepare Delivery Note with RL/BL strict separation



15.1 Pages visible to Warehouse User

- Live Dashboard
- Warehouse Portal (this page)
- Reports
- Locator - which rack a material is on, and the reverse: scan a rack and see everything that should be on it, which turns a stock count into a checklist (see §21.8)
- Assets - serialised tools and equipment, each with its own record, status and location history, including a map link where a position was captured (see §21.7)

15.2 Sidebar warehouse resolution

- A `warehouse_user` is bound to a single warehouse via `users.Warehouse_ID`. The portal auto-resolves from the user's profile.
- An admin shadowing the portal gets a sidebar "Shadow warehouse" selectbox listing every active warehouse.
- If neither resolves: red error card "Your account is not bound to a warehouse. Ask Admin to set your `Warehouse_ID` in Admin Portal -> Users."

15.3 Page title

Warehouse <ID> in gold, with (prices hidden - Logistics-only) muted caption next to it as a permanent visual reminder.

15.4 Tab 1: Incoming Assignments

POs Logistics has routed to this warehouse.

15.4.1 Hero strip

Card	Source
AWAITING ACK	Count of assignments with status='assigned'
ACKED	Status = acknowledged (you've seen them, waiting on goods)
IN/RECEIVED	Status in partial / received

15.4.2 Assignments grid

A table with columns: Assign # - PO No. - PR No. - Vendor - Dest Site - Expected - Assigned - Acked - Status. No price columns.

15.4.3 Acknowledge action

Selectbox of status='assigned' rows. Picking one -> [OK] Acknowledge button -> flips to acknowledged and pings Logistics with warehouse_acknowledged notification (off by default in WhatsApp toggles to keep noise low).

15.5 Tab 2: Receive Goods

Record qty actually received at this warehouse against an acknowledged assignment.

15.5.1 Assignment picker

Selectbox of status IN ('acknowledged', 'partial') rows.

15.5.2 PO snapshot card (no prices)

KV row	Value
PO Number	the PO
Vendor	<code> - <name>
Inco / Payment	the terms
Expected	the date

Total_Amount, Freight_Charges, Handling_Charges, Discount_Amount, Amount_In_Words are stripped from the header dict before render. Never visible.

15.5.3 Receive grid

Editable grid showing every line on the assignment:

Column	Source
id	po_items.id (disabled - read-only)
Material_Code	line code
Description	line description
UOM	unit
rl_bl_family	family chip (RL / BL / blank)

Column	Source
Qty	ordered (read-only)
Delivered_Qty	already received cumulative (read-only)
Open_Qty	computed Qty - Delivered + Returned (read-only)
Receive Now	EDITABLE - type the qty you physically received this event

15.5.4 Record receipt button

- Validates: at least one row with Receive Now > 0
- Over-deliver guard: Delivered_Qty - Returned_Qty + new_qty may NOT exceed ordered Qty. If it does, the entire batch is rejected with a friendly message naming the line.
- On success: bumps each line's Delivered_Qty, flips line_status (delivered if Delivered - Returned >= Qty, else partially_delivered)
- Rolls assignment status (received if every line on the parent PO is delivered, else partial)
- Rolls PO header status to match
- Notification: warehouse_received to Logistics with line count

15.6 Tab 3: Prepare DN

Build a Delivery Note for a site. RL/BL strict separation is enforced here - a DN cannot span both families. If you try, the action is rejected with: "Strict separation violated: this DN spans multiple RL/BL families. Prepare one DN per family."

15.6.1 Source pickers

- PO No. selectbox (assignments with status acknowledged / partial / received)
- Destination Site selectbox

15.6.2 Items grid

Editable grid with read-only inventory columns (Material_Code, Description, UOM, rl_bl_family, Qty, Delivered_Qty, Returned_Qty) and editable shipping columns:

Column	Purpose
Ship Qty	What you're sending on this DN (must be > 0 to include)
Lot_Number	If you tracked the lot at receive time
Expiry_Date	Lot expiry
Remarks	Free text

15.6.3 DN header

Three-column row: DN Date - Vehicle No - Driver Name - Driver Phone - Prepared By (auto-filled with your username) - Remarks.

15.6.4 Save DN draft button

- Validates: at least one row with Ship Qty > 0
- Over-ship guard: available = Delivered_Qty - Returned_Qty - Sum(live DN qty) per line; ship qty

cannot exceed available

- RL/BL strict-separation check: if the items span both families, REJECTED
- On success: new delivery_notes row with status='draft' and one dn_items row per line. The DN header carries rl_bl_family if non-NULL.
- Toast: DN <number>

15.6.5 DN numbering convention

DN-<WAREHOUSE_ID>-<YYYYMMDD>-<seq> - seq resets per (warehouse, day). Example: DN-WH-A-20260616-003.

15.7 Tab 4: Outbound DNs

Track every DN this warehouse has prepared.

15.7.1 Status filter

Multi-select of every DN status. Defaults to draft, pending_logistics, pending_hod, pending_sk (the active ones).

15.7.2 DN grid

A table of every matching DN with full state metadata.

15.7.3 Submit-to-Logistics action

When the filter shows any draft row, a "Submit a draft to Logistics" section appears with a selectbox of drafts + Submit to Logistics button. Submission flips status to pending_logistics and pings Logistics with the DN-approval-queue notification.

15.7.4 Per-DN drilldown

Select a DN -> section card showing PO, Site, Status, RL/BL family, Vehicle/Driver, and the full signature trail (Logistics decided by/decision, HOD decided by, SK received by). Items grid below.

15.7.5 Request reschedule expander

If the date you targeted isn't going to work, raise a reschedule from here. Same flow as the HOD's In-Transit tab (defaults, min_value=today, mandatory reason).

15.8 Tab 5: <- Returns from Site

Raise a return when a site flags defective material from a DN this warehouse delivered.

15.8.1 DN picker

Selectbox of DNs in received / pending_sk / hod_approved states.

15.8.2 Items grid

Editable grid with a Return Qty column. Filling > 0 on a line includes it in the return.

15.8.3 Reason textarea

Mandatory.

15.8.4 <- Raise return to vendor button

- Internally calls the system which fans out to the system per affected line
- Each line: po_returns row written with raised_by_role='warehouse_user', dn_item flagged returned, parent po_item's Returned_Qty bumps and line_status flips back to partially_delivered or open
- PO header reopens if it had been closed
- Notification fires to Logistics

15.9 Tab 6: History

Read-only. Two sub-tabs: Completed DNs (status in received / rejected / cancelled) and Closed assignments.

15.10 Warehouse - Use Cases

Use Case 1: Receive a vendor delivery

1. Incoming Assignments -> [OK] Acknowledge the assignment when you see it
2. Goods physically arrive
3. Receive Goods -> pick the assignment -> type the actual received qty per line -> Record receipt

Use Case 2: Ship to a site (single family)

1. Prepare DN -> pick PO + destination site
2. Type Ship Qty per line - keep all RL or all BL
3. Fill DN header (Vehicle, Driver, Date)
4. Save DN draft -> status = draft
5. Outbound DNs -> select the draft -> Submit to Logistics

Use Case 3: Ship RL + BL to the same site on the same day

You need TWO DNs:

1. Prepare DN with only RL lines -> save + submit
2. Prepare second DN with only BL lines -> save + submit

The system rejects any attempt to combine them on a single DN.

Use Case 4: A site rejects part of a DN - raise to vendor

1.

- <- Returns from Site -> pick the DN
- 2. Type Return Qty on the offending line
- 3. Reason: "Site reported defective surface coating"
- 4. <- Raise return to vendor - PO + po_item reopen; Logistics sees the return in their tab

Use Case 5: Request a reschedule

- 1. Outbound DNs -> drill into the affected DN
- 2. Request reschedule for this DN -> pick new date + reason -> Submit

15.11 Warehouse - FAQ

Q: Why don't I see prices anywhere? A: Warehouse role is intentionally blind to commercial data. Three independent enforcement layers strip every monetary field. If you genuinely need a price (e.g. to file a damage claim), ask Logistics or Admin.

Q: My over-ship guard rejected a DN. I just received everything from the vendor. A: Probably another DN is already drafted (or submitted) shipping a slice of that line. Check Outbound DNs filter for live DNs against the same PO. If one was abandoned, ask Logistics to reject/cancel it so the qty frees back up.

Q: RL and BL are going to the same site on the same truck. Can I combine? A: No. Two DNs. The strict separation is by design (different testing standards, different storage requirements). The site receives them as two separate DNs in their Receipt Staging queue.

Q: Where did the "Receive Goods" assignment go? It was here this morning. A: Once every line on the parent PO is fully delivered, the assignment moves from partial to received and stops appearing in Tab 2 (which filters to acknowledged + partial). Check the assignment grid in Tab 1.

Q: Can I edit a DN after submitting to Logistics? A: Not directly. Ask Logistics to reject it (Outbound DNs shows the rejection if it happens) - then it returns to draft for you to edit.

16. Cross-Role Procurement Walk-through

A single happy-path narrative threading all five roles together, useful for onboarding.

16.1 The scenario

Site GI-PS01 needs 50 RL panels and 20 BL bricks. Total 3 working days from PR raise to physical receipt.

16.2 Hour by hour

When	Who	What
Day 1 - 8:00 AM	Site HOD	Opens HOD Portal -> Purchase Requests. Creates 2 PR lines (50 RL panels, 20 BL bricks) with PR Number 3000099999, WBS Number 4003951, Network 4003951-PROJ-A, Plant
Day 1 - 8:15 AM	Site HOD	Opens the Delivery PRs Day 1 Logistics Portal expander -> multi-selects PR 3000099999 -> Submit Selected to Logistics.
Day 1 - 8:15 AM	(auto)	pr_submitted_to_logistics notification -> Logistics inbox (red badge). WhatsApp ping if enabled.
Day 1 - 9:00 AM	Logistics	Sees PR 3000099999 in Incoming PRs queue. Drills in - 2 lines, RL + BL.
Day 1 - 9:30 AM	Logistics	Switches to Create PO -> manual entry. Picks vendor "Carborundum Universal" (0000110341), Inco/Payment auto-fill, fills PO Number 4720033030, Unit_Price per line,
Day 1 - 9:30 AM	(auto)	PO auto Price computed. Save PO. logistics_status='in_po' -> disappear from Logistics queue. Site HOD gets po_issued notification
Day 1 - 10:00 AM	Logistics	Assign to Warehouse -> picks PO 4720033030 + Warehouse WH-A + Expected Delivery Day 2 evening. Assign.

Day 1 - 10:00 AM	(auto)	po_assigned_to_warehouse notification -> WH-A users' inbox. WhatsApp ping if enabled.
Day 1 - 11:00 AM	Warehouse user (WH-A)	Incoming Assignments -> [OK] Acknowledge assignment #N. Goes about their day.
Day 2 - 5:00 PM	Warehouse user	Physical truck arrives from vendor. Receive Goods -> assignment #N -> type 50 in Receive Now for the RL line, 20 for the BL line. Record receipt.
Day 2 - 5:30 PM	Warehouse user	Prepare DN -> pick PO 4720033030, destination GI-PS01. Types Ship Qty 50 for RL line, attempts to add BL line - rejected (RL/BL strict separation). Saves DN-WH-A-20260617-001 for
Day 2 - 5:35 PM	Warehouse user	Repeats: new DN with only the BL line.
Day 2 - 5:40 PM	Warehouse user	Outbound DNs -> submits both drafts to Logistics.
Day 2 - 5:45 PM	Logistics	Sees DN-WH-A-20260617-001 + 002 in their DN approval queue. Confirms delivery date Day 3 AM. Approves both.
Day 2 - 5:45 PM	(auto)	DN status flips to pending_hod. Site HOD gets dn_logistics_approved notification.
Day 2 - 6:00 PM	Site HOD	Logs in, sees red bell badge 2 unread. Opens inbox -> drills into the DNs via
Day 2 - 6:00 PM	(auto)	DN status pending_sk. Two mirror rows in pending_receipts.status='pending_sk' . Site SK gets dn_auto_generated notification.
Day 3 - 8:30 AM	Truck arrives at site GI-PS01	
Day 3 - 8:45 AM	Site SK	Opens Entry Log -> Receipt Staging. The new Incoming Delivery Notes from Warehouse expander is open at the top, showing both DNs. Inspects both, confirms physical match, clicks [OK] Mark as Received on each.

Day 3 - 8:45 AM	(auto)	Two receipts rows written (with DN_Number, Warehouse_ID, PO_Number_Source). Both DNs flip to received. pending_receipts mirror rows deleted. Inventory cache busted. Live Dashboard shows the new stock immediately.
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16.3 Side-path examples

A site rejects a DN

Same scenario, but on Day 2 at 6:00 PM, Site HOD inspects the line preview on DN-002 and notices the BL qty doesn't match the PR. Rejects DN-002 with note "BL qty in DN is 25, PR asked for 20 - bounce back".

- DN-002 status -> rejected, rejection_reason stored
- Warehouse user gets the rejection notification with the reason
- The BL line on the PO frees up qty (the over-ship guard recalculates available since DN-002 is no longer "live")
- Warehouse prepares DN-003 with correct qty

Logistics force-closes the PR mid-flight

Project cancelled by management on Day 1 evening:

- Logistics -> Force-Close -> radio = PR -> reason "Project cancelled per CFO email 2026-06-16" -> Force-close PR
- Admin + Site HOD see critical-severity notification in their bell
- The pending PO can either stay (if vendor already shipped) or also be force-closed
- Site HOD's In-Transit -> Force-closures sub-tab shows the closure with the reason

Warehouse asks for reschedule

Receiving day Truck doesn't show up - vendor delay:

- Warehouse -> Outbound DNs -> drill into DN-001 -> Request reschedule -> new date Day 4 -> reason "Vendor delivery delayed 24h confirmed by vendor email"
- Logistics -> Reschedules -> reviews -> [OK] Approve with notes "Confirmed with vendor"
- PO Expected_Delivery, po_assignments.Expected_Delivery, AND DN_Date all flip to Day 4 in one transaction
- Site HOD sees the decision notification

17. Operations & Hosting - the after-launch chapter

This chapter is for whoever is responsible for keeping the system running once it is live. Everything before it describes how to use the application; this chapter describes how it is hosted, what protects it, and what to do when something goes wrong.

It is written for a manager, not an engineer. Nothing here asks you to type a command. Where a task genuinely needs technical work, this chapter says what the task is, who performs it and how long it takes. The step-by-step procedures live in the technical runbooks listed in section 17.8, kept separately for whoever administers the server.

17.1 How the system is put together

Four layers sit between a user and your data, and each has a single job.

Layer	What it is	What it does for you
1. The user's device	A phone, tablet or laptop - browser or installed app	Where work gets entered. No company data is stored on the device
2. The security edge	A commercial protection service sitting in front of the system	Checks the person's company email address before they reach the login page, absorbs attacks, and renews the
3. The application server	The server that runs GI Hub	Runs the application, holds the database, sends the notifications
4. Nightly backup	An encrypted copy held off the server	A second copy of everything, taken every night, kept for 14 days

Two properties of this arrangement are worth understanding, because they are what most of the security answers in section 17.5 rest on:

- The connection runs outward, not inward. The server opens a connection to the security edge; the edge never opens one to the server. There is no open door for anyone to find - someone scanning the internet for our system sees nothing at all.
- Your data stays on your own server. The security edge forwards encrypted traffic it cannot read. The backup copy is encrypted too. No outside party holds a readable copy of your stock figures, prices or documents.

17.2 Bringing a new installation live

Standing up a fresh installation is a one-off technical job for whoever administers the server. It is listed here so you know what it involves, what to ask for, and what must be true before the address is shared with staff.

#	Step	Who	Roughly
1	Install the application and its	Administrator	30 min
2	Services on the server Connect the server to the security edge and point the company web address at it	Administrator	15 min
3	Set the email allow-list so only company addresses can reach the login page	Administrator	10 min
4	Connect the WhatsApp notification channel	Administrator	10 min
5	Complete the hardening checklist below	Administrator	15 min
6	Take a first backup and confirm it landed	Administrator	5 min
7	Change every default password in the application	You	10 min

Before the address is shared with anyone

These five are not optional. Each one closes a specific, realistic way of losing control of the system.

1. Disk encryption is switched on on the server. A stolen or salvaged disk is then unreadable rather than a complete copy of your business.
2. Every default password has been changed. The accounts the system creates on first start are publicly documented in this manual - see section 2.4.
3. The email allow-list is in place. Until it is, the login page is visible to the whole internet.
4. The server does not sleep or lock. A sleeping server is an outage.
5. A first backup exists and has been checked. An untested backup is not a backup.

17.3 Day-to-day running

In normal operation there is nothing to do. The system runs as a set of background services that restart themselves if they stop, and the backup runs overnight without being asked.

Four routine needs come up, and all four are one request to your administrator:

What you need	What to ask for	Impact on users
Confirm everything is healthy	A service status check	None
Investigate a reported problem	The recent activity logs	None
An immediate extra backup	A one-off backup, before risky work	None
Apply a new version	An application restart after the update	About 3 seconds

Applying an update

Updates are additive and reversible, which is why the downtime is measured in seconds rather than minutes:

1.

The administrator applies the new version.

2. The application restarts. Anyone mid-form keeps their work - they may see a brief reconnect.
3. Any new database structure the version needs is applied automatically before the first person can sign in. Nobody runs a database update by hand.

Rolling back. If a release misbehaves, the administrator can return to the previous version in about a minute. Your data is unaffected: the system only ever adds to its database structure and never removes anything, so an older version reads newer data perfectly well. You do not lose the work entered between the upgrade and the rollback.

Take a backup first when

- A major version upgrade is being applied.
- Bulk edits are about to be made through the Admin Portal's data editor.
- The system is moving to different hardware.

17.4 Security and data safety - what to tell management

These are the questions that come up in a management or client review, with answers you can give directly.

Where does our data physically live?

- On our own server. The database and every uploaded attachment - delivery notes, material certificates, photographs - sit on storage we control. No third-party service holds a copy of your inventory.
- The disk is encrypted. A stolen or salvaged disk is unreadable without the key, rather than a complete copy of the business.
- The nightly backup is encrypted too, and is held somewhere the server itself cannot reach - so a problem on the server cannot corrupt the backup.

Who can reach the application?

There are two locks on the door, not one:

1. The security edge checks the email address first. Anyone not on the allow-list never reaches our server, never sees the login page, and leaves no trace on our system at all.
2. The application then requires a username and password. Passwords are stored using a one-way scramble, so even someone holding the database file cannot read them back.

Sensitive accounts can add a second factor, and a session that sits idle signs itself out rather than waiting to be found.

How is the connection secured?

- Every byte travels encrypted, using the same standard as online banking.
- The security edge forwards encrypted traffic and re-encrypts it on the way to our server. The contents of your records are not readable by it.

-

The provider holds the recognised independent certifications for this class of service - SOC 2 Type II, ISO 27001 and PCI DSS.

What is the attack surface?

- No open ports. Our server never accepts an incoming connection from the internet; it makes an outgoing one instead. Someone scanning for our system finds nothing to attempt.
- Every action is permission-checked on the server, not just hidden in the interface. A user cannot reach another role's function by guessing an address, and a read-only account is refused every change centrally rather than screen by screen.
- Every consequential action is recorded with the user's name, the time and what changed. Admin can review or export that history at any point.

What happens if something fails?

Failure	Impact	Recovery
Power loss at the server	System offline while it lasts	A battery unit covers short cuts; otherwise it restarts by itself when power returns
Internet outage	Remote users cannot reach it; staff on	None needed - it resumes on its own
Security-edge provider incident	Remote access blocked	An alternative route can be switched on in about 5 minutes
Server hardware failure	Offline until replaced	Restore the last backup onto replacement hardware: about 30 minutes, losing at most one day
Disk corruption	Database unusable	Restore from any of the last 14 nightly
A record deleted by mistake	One record lost	Restore about 5 minutes Restore from the live system and re-enter it
A member of staff misuses their access	Limited to what their role can reach	The audit trail names the user, the action and the exact time

The two rows worth acting on before you need them: a battery unit for the server removes the most common cause of an outage, and a second backup destination (section 17.5) removes the only single point of failure left in the recovery plan.

What does it cost to run?

Running costs are dominated by the server itself; everything around it is either free at this scale or a few dollars a year.

Item	Cost
Web address	About \$8 a year
Secure connection to the server	Free, unlimited traffic
Email allow-list, up to 50 users	Free
Backup storage at this data volume	Free

Item	Cost
WhatsApp and email notifications	Included
AI assistant	Free - it runs on our own server, so no per-question charge
Server and electricity	The main line item, and it depends on the hosting choice

Non-negotiables

Skipping any one of these defeats the security model rather than weakening it.

1. Disk encryption stays on.
2. The server stays in a physically secured location.
3. Default passwords are changed on day one - they are published in this manual, in section 2.4.
4. The email allow-list is in place before the address is shared.
5. The server is not used for anything else. Keep it single-purpose.
6. Operating-system security updates are applied within a week of release.

17.5 Backup and restore

What runs automatically

Every night at 02:00 the system takes a complete copy of itself, without going offline and without anyone starting it:

1. A consistent snapshot of the database - taken safely even while people are using the system, so there is no maintenance window.
2. A copy of every uploaded attachment: delivery notes, material certificates, photographs.
3. Snapshots older than 14 days are removed, so the backup area does not grow without limit.
4. Optionally a second copy to another destination, if one has been configured.

Restoring

Restoring is a technical procedure your administrator performs. What matters to you is the shape of it:

Question	Answer
How far back can we go?	Any night in the last 14 days
How long does a restore take?	About 5 minutes for the database; a little longer if attachments are also being restored
Is the system usable during a restore?	No - it is stopped for those few minutes, because the database must not be open while it is replaced
How much work would we lose?	Everything entered since the snapshot being restored. Restoring last night's copy loses today
Can a single deleted record be restored on its own?	Not directly. The practical route is to restore a copy alongside the live system, read the record out of it, and re-enter it

Where the copies live

One nightly copy goes to encrypted cloud storage. That is one copy in one company's hands, which is better than none but is not two independent failure domains.

If the data matters enough to survive a bad week - and inventory, procurement and valuation records normally do - ask your administrator to add a second destination in a different place. Any of these work, and all of them are cheap:

Option	Cost	Set-up
A second cloud storage provider	Free at this data volume	About 15 minutes, one-off
An external drive kept on site	Cost of the drive	Minutes, one-off
A drive held off site, rotated	Cost of two drives	Minutes, plus the discipline to rotate them

The recommended arrangement is the nightly cloud copy plus one other provider: three independent things would then have to fail on the same night before data is genuinely gone.

17.6 Controlling who can reach the system

Access is controlled in two independent places, and it is worth being clear about which does what:

Control	What it decides	Who changes it
The email allow-list, at the security edge	Whether a person reaches the login page at all	Administrator, in the security dashboard
The user account, inside the application	What that person can see and do once signed in	You, in Admin Portal -> Users

Both must be right. An allow-listed email with no user account cannot sign in; a user account whose email is not allow-listed never sees the login page.

Adding or removing someone

When a new person joins, two things must happen: their company email is added to the allow-list, and you create their account with the correct role and site. Neither works without the other.

When someone leaves, do both, in this order:

1. Deactivate their account in Admin Portal -> Users. This takes effect immediately and is the one that matters - it ends any session they have open.
2. Ask for their email to be removed from the allow-list. This takes effect globally within about a minute.

Changing the allow-list needs no restart and interrupts nobody.

A note on rate limiting

Repeated failed sign-in attempts are already slowed down by the application itself. Ask your administrator to add a limit at the security edge as well - it stops that traffic before it ever reaches your server, which is both faster and cheaper than handling it inside the application.

17.7 When something is not working

Find the symptom, try the first column, and escalate if it persists. None of these need you to type anything.

What you are seeing	Most likely cause	What to do
The web address shows an error page instead of the login screen	The application or its connection to the security edge has stopped	Ask your administrator for a service status check. This is almost always a restart
A valid company email is refused at the login page	That address is not on the allow-list	Ask for it to be added - takes about a minute to take effect
Someone signs in but sees "your account has no site"	The account was created without a site	Admin Portal -> Users -> set their site (or warehouse, for warehouse staff)
The application works but WhatsApp messages are not arriving	The notification assigned has become disconnected	Ask your administrator to re-connect it. Nothing is lost - messages queue
Notifications arrive late in a batch rather than immediately	The recipient has the evening-summary preference switched	That is intentional. Change it in their profile if they want them immediately
The AI assistant says it is offline	The assistant service is not running	Ask your administrator to start it. Every
Times on records look wrong by a fixed number of hours	The server's time zone is not set correctly	Ask your administrator to correct it and
Backups have stopped appearing	The backup destination is full or disconnected	Ask for the backup log to be checked. Treat this as urgent - you are running without a safety net

Escalating to the developer

Three things make a report actionable, and without them the first reply will only ask for them:

1. What you expected to happen, and what happened instead.
2. The exact time, and which site and user account it happened on.
3. A screenshot, including any error text in full.

Your administrator can attach the relevant service logs. Please do not delete or re-enter the affected record before it has been looked at - the audit trail is usually what identifies the cause.

17.8 Technical runbooks

The procedures behind this chapter are documented for whoever administers the server. They are deliberately kept out of this manual, which is written for people who use the system rather than run it.

Runbook	Covers
Deployment guide	Server setup, the security edge, going live
Debugging guide	Diagnosing faults, reading the logs
Native apps guide	Building and distributing the Windows, macOS and Android
Architecture reference	How the system is built, and why

Runbook	Covers
Standard operating procedure	Day-to-day operating rules for every role

18. Material Estimator (SME) Manual

> Access: Material Estimator page - exact-locked to HOD + Admin. HOD is scoped to their own site; Admin gets a sidebar site picker. SK / Supervisor / Logistics / Warehouse never see this page.

18.1 What it is

The Smart Material Estimator (SME) is a planning portal for Rubber-Lining / Brick-Lining work. It answers: "For the equipment and lining systems at this site, how much material do we need, what's already available, and what must we still procure?" It is read-only - planning here never writes a consumption, a receipt or a stock figure.

> The estimator is a separate pool from the warehouse, deliberately. > Every quantity the estimator uses comes from its own material baseline and from nowhere else. A warehouse issue does not reduce what the estimator says is available, and it is not supposed to: the plan answers "what does this project need and what has been bought for it", while the warehouse answers "what is on the shelf today". Netting one off the other would silently move every readiness figure on every tab. Where physical draw matters, it is shown beside the plan as an Actual Physical Balance - adjacent, never merged. See section 21.

It works from three sets of master data, all maintained on the Master Data tab:

Master data	Scope	What it holds
Equipment	Per site	Each tag, its location, its surface area and which lining system applies
Recipe	Shared	How much of each material one square metre of each lining system needs
Materials	Shared	The estimator's own quantity baseline - what has arrived and what is still on order

Locations and equipment types come from the system settings, so the drop-downs stay consistent across every tab.

18.2 Tabs (8)

Tab	Purpose
Dashboard	KPI strip + per-location material balance + a stock-only view of SME-tracked items + a procurement view (what to buy, per code/location).
Priority	Drag-to-rank equipment so the allocation engine fills the highest-priority tags first when stock is short.

Session / Selective Order	Pick specific equipment, enter the SQM to execute, and get the exact material list + a combined procurement total with
Location Report	Material Report Material broken down by location, with per-location colour schemes; plus an "All Equipment"
Equipment Report	Per-equipment three-section breakdown (requirement / available / shortfall), exportable as a multi-sheet
Execution Plan	Orderable items card + procurement-priority + production-detail blocks for executing a work package.
Total Overview	Whole-site rollup: 6-card KPI strip + master table + per-system-code drill-downs.
Master Data	CRUD for Equipment, Recipe, and the SME Materials baseline + Locations/Types. Writes go only to the SME-owned tables - never the ERP inventory ledger.

18.3 How SME consumption reaches the ledger

The SME does not issue stock itself. Day-to-day consumption is entered on the Store Keeper -> SK Consumption page (the SME Multi-Material Entry grid): pick an equipment tag -> system codes -> SQM per system -> auto-computed materials -> submit a batch. That flows through the normal EOD commit pipeline (HOD approval -> commit_eod), and the SME's Done_SQM advances on commit. So the estimator's "available" figures stay in sync with real ledger movements.

18.4 Key rules (for support)

- SME inventory is isolated from ERP inventory. The SME baseline lives in its own seed table; quantities shown are derived, not stored. Do not expect SME edits to change the Live Dashboard stock and vice-versa.
- Master-data grids order by an explicit key (not row order); if a grid says "No records found" but data exists, it's a known historical bug class - refresh / report it.
- Bootstrap/seed is loaded by Admin via scripts/sme_bootstrap.py --site-id <SITE>.

19. Man-Hours & Labor Tracking Manual

> Access: Man-Hours page - exact-locked to HOD + Admin. HOD is scoped to their own site; Admin gets a sidebar site picker. (NEW - 2026-06.)

19.1 What it is

Man-Hours tracks labor the way the Material Estimator tracks material: log who worked, where, and for how long; record SQM produced; define the required man-hours for a scope; then compare estimated vs actual to surface over-runs. It is fully site-scoped and isolated - it reads the Material Estimator's Equipment / System-Code / Location lists (read-only) for its dropdowns, and writes only to its own mh_* tables. Nothing here touches inventory, the EOD path, or the SME data.

19.2 Tabs (5)

19.2.1 Employees

Per-site labor roster. Add/update a worker with Code, Name, Designation, Type (OWN / Supply), Company. "OWN" = your own (GI) staff; "Supply" = subcontractor labor (e.g. DMC) - these can live here even though they aren't ERP app-users. A status expander flips a worker active/inactive.

19.2.2 Daily Timesheet

Two ways to enter actuals, plus team-SQM:

- Excel upload - upload an attendance workbook in the to_john_Attendance format (sheets ADD EMPLOYEE + SAR). The system previews the parsed counts, then on import you choose:
- Replace rows for these dates (recommended) - deletes this site's existing timesheets for the dates in the file, then inserts (safe to re-upload a corrected file - no duplicates), or
- Append - adds the rows as-is. Unknown workers are auto-created into the roster. Location / Equipment / System import blank (the source file doesn't carry them) and are assigned afterward here.
- Manual per-day batch grid - pick a Work Date, Equipment Tag, System Code (drop-downs from the Material Estimator's equipment), and a break (default 60 min). The grid lists active workers; tick Worked and set In/Out (default 07:30-16:30). On save, one timesheet row is written per worker.
- Team SQM - record the SQM the crew completed on that date/tag/system and choose even (equal per worker) or by-hours (pro-rata on hours) distribution; each worker's share lands in their row's Allocated_SQM.

Hours math: Total = (Out - In) - break; Normal = min(Total, 8 h); OT = max(0, Total - 8 h). (Example: 07:30-16:30 with a 1 h break = 8 h Total / 8 Normal / 0 OT.) The source file's own hour columns are ignored - hours are always recomputed.

19.2.3 Estimator

Define the required man-hours for a scope: pick Equipment Tag + System Code, enter Estimated man-hours (and optionally Estimated SQM -> yields a man-hours-per-SQM norm) + a basis note. One estimate per Tag/System (re-saving updates it).

19.2.4 Estimate vs Actual

The variance dashboard. KPI strip (scopes tracked / over-consuming / total actual man-hours) + a table where over-runs >10 % are highlighted red and on-or-under-budget rows green. Variance = Actual - Estimated, Variance % = Variance / Estimated. A "Where the most man-hours went" table ranks the top consumers. An expander lets you record a reason for an over-consumption (saved per Tag/System and shown back in the table).

19.2.5 Employee-wise

The "where did each person work, date by date" view. Pick an employee (or All) and a date range; get a clean, date-ordered list of every tag/system they worked with hours and allocated SQM - the un-cluttered per-person timeline.

19.3 Loading an attendance file in bulk

Attendance is normally uploaded through the Man-Hours page itself, which is the recommended route: it shows you what it parsed before anything is saved.

For a very large historical backfill, your administrator can load the same file directly on the server. It uses the identical parsing and import logic as the page, so the result is the same either way, and it offers two options worth knowing about:

Option	What it does	When to ask for it
Preview only	Parses the file and reports what it found, saving nothing	Always, the first time a new file format
Replace by date	Existing lines for the dates in the file are replaced	Replacing a corrected file for a period
Add without replacing	Lines are added alongside what is already there	Loading a period that was never loaded before

Replace by date is the default, because loading a corrected file twice is a far more common mistake than deliberately wanting two copies of the same day.

19.4 Key rules (for support)

- Isolated mh_* data. Man-Hours never changes inventory, SME data, or the EOD ledger.
- Site-scoped. HOD sees only their site; Admin switches sites with the sidebar picker. Each timesheet/estimate carries Site_ID.
- One timesheet line per (employee, date, equipment-tag, system-code). A worker split across two tags in a day has two rows.
- Equipment-Tag / System-Code / Location drop-downs come from the Material Estimator's equipment master - keep that seeded for a site or the lists will be empty.

20. Auditor (View-Only) Manual

Added 2026-08-03.

20.1 What the Auditor role is for

The Auditor is for anyone who must see the numbers without being able to move them - an internal auditor, a client-side QA reviewer, a finance reviewer, a new manager still learning the system, or a consultant given temporary access.

It is a real role like any other, with one difference that is absolute:

> An Auditor can read everything its level reaches and can change nothing, > anywhere, ever.

That is not a UI convention. It is enforced on the server for every request.

20.2 What an Auditor can see

Area	What it shows
Dashboard	Every KPI, chart and category breakdown, across all sites
Stock	Current stock by site, minimum levels, expiry
Records	Inventory, Receipts, Consumption, Returns, Lots, Purchase
Reports	Every report, user Requests as Excel, PDF or CSV
Lining Coverage	The read-only coverage analysis
Documents	The SOP and User Manual PDFs, QR/label sheets
Account	Own security settings (password-manager 2FA, phone number)

An Auditor is not site-locked. Every other role below Logistics sees only its own site; an auditor sees all of them, because an auditor confined to one site cannot audit the business.

20.3 What an Auditor cannot do

Everything that changes data. The Data Entry forms, Master Data editors, Bulk Excel Import, HOD Approvals, the Logistics and Warehouse portals, the Supervisor portal and the Admin console do not appear in the sidebar at all - they are not hidden buttons on a visible page, the pages are simply not reachable.

On pages an Auditor can open, actions that would change something are disabled rather than hidden, with a tooltip explaining why:

> Your account is view-only (Auditor) - this action changes data.

On the Reports page, for example, Excel / PDF / CSV downloads work normally (they only read), while Archive, WhatsApp, New schedule, Run now and Delete are greyed out.

A "View only" tag sits permanently in the header beside the user name, so there is never any doubt about which kind of account is signed in.

20.4 What an Auditor CAN still do

Because they are read-only, not crippled:

- Download every report in all three formats.
- Use the CmdK / Ctrl-K command palette to jump to any page or material.
- Ask the Hub Assistant questions (it answers from the Auditor's own manual chapters - see §20.6).
- Manage their own account: enable or disable two-factor authentication and change their own phone number. These touch only their own user record.

20.5 The security model (for Admins)

Worth understanding before you hand the role out.

The restriction is enforced once, centrally, rather than screen by screen. Any request that would change data from a view-only account is refused before it reaches the feature at all, and the user is told plainly that their account is view-only and the action is not permitted.

Why that design matters to you: a per-endpoint permission check is only as reliable as the developer who remembers to add it, and the one that gets forgotten fails open - the write succeeds and nothing looks wrong. Keying on the method inverts that, so a feature added next year is closed to Auditors from the moment it is written. 126 of the system's 143 state-changing endpoints are blocked; the 17 that are not are sign-in, sign-out, token refresh, the user's own 2FA and phone-change flows, and read-only calculation endpoints that use POST only because the request is too large for a URL.

Two consequences to be aware of:

- An Auditor probing a URL that does not exist gets 403, not 404. The guard runs before routing, so it cannot be used to map the system.
- Blocking is by role, not by page. Even if an Auditor somehow reached a write page, every button on it would still fail server-side.

20.6 Creating an Auditor account

Admin Portal -> Users -> Add user, and pick Auditor (view-only) in the Role dropdown. Leave the Site blank - Auditors are global by design, and the form does not ask for one.

Users can also request the role themselves on the registration screen; it still needs Admin approval like any other request.

20.7 The Hub Assistant for Auditors

The assistant answers from chapters 1, 2, 3, 8, 9, 10, 11, 12, 16 and 20 - orientation, the reports module, notifications, the data model, the glossary, the FAQ and the cross-role procurement walk-through. That is

deliberately the set matching the pages an Auditor can open.

Ask it about a report, a status code or what a field means and it will answer. Ask it how to post a consumption entry and it will tell you that is outside the read-only Auditor view - because the operational chapters for roles that can post entries are never loaded into an Auditor's context in the first place.

21. 2026-08 Feature Update - What Changed

Everything in this chapter is live. Grouped by what you would notice.

21.1 Reports and exports look the same everywhere now

Every Excel export - not just the Material Estimator's - now carries the branded layout: the GI logo, a "Report Generated / Generated By" band, a title bar and a framed, filterable table with the header frozen. Previously only three SME workbooks looked like that and everything else came out as a bare grid.

> If you have automation reading these files: the header row is now row 6 > and data begins on row 7 (rows 1-4 are the logo and meta band, row 5 the > title bar). Anything that assumed a header on row 1 needs adjusting.

PDF reports no longer overlap or cut off text. Columns are now measured against their actual content and share the page fairly, and long values wrap onto more lines instead of being truncated. Previously every column got an equal slice of the page regardless of need, so a long material description was drawn on top of the next column while Date and UOM wasted most of theirs - and anything past 24 characters was silently deleted.

QR badge and bin labels were fixed the same way: a long employee name used to run past the edge of its sticker onto the one beside it.

21.2 The Material Estimator understands your order data correctly

Ordered_Qty is the TOTAL quantity procured for the project, and Available_Qty is the part of it that has already arrived - available is a subset of ordered, not a separate pile beside it.

The estimator used to add the two together, which double-counted every unit already on the shelf. On real data that understated the buy list by 22,951 units across 22 of 30 materials, and on GI-8005763 - where all 143,000 ordered units had arrived - it read 286,000 and reported nothing to buy against a demand of 152,685, hiding a 9,685-unit shortage completely.

What you will see in the UI:

- Second-tier quantities now read "Pending Delivery" - the part of the order still on its way - never just "Ordered".
- Second-tier coverage reads "When delivered".
- Reports carry a Total Procured column beside them.

Feasibility is unchanged and still judges physical stock only: a tank cannot be built with a purchase order.

21.3 A view-only Auditor role

See §20.

21.4 You are signed out after 30 minutes of inactivity

Leave the app untouched for 30 minutes and it signs you out and returns you to the login screen. A warning appears two minutes before with a button to stay signed in, and any real activity - a click, a keystroke, a scroll - resets the clock silently.

This matters most on shared site terminals, where an unattended session was previously good until the browser was closed.

21.5 Search anything with CmdK

Press CmdK (Mac) or Ctrl-K (Windows) anywhere in the app.

The palette already jumped to pages; it now also searches your inventory live. Type a SAP code, part of a material name or a material code and the matching items appear under a Materials heading - pick one and you land on its Material Intelligence card. Results are scoped to what your role may see.

21.6 The Hub Assistant is faster and more accurate

The assistant used to be handed its entire allowed portion of this manual on every single question. For an Admin that was the whole ~180 KB document, which is slow to process; for everyone else it was the first 800 characters of each chapter, which is the wrong 800 whenever the answer is further down.

It now finds the passages that actually match your question and reads only those - a 97.7 % smaller prompt for Admins, and noticeably better answers for everyone, because the relevant paragraph is now present in full instead of being cut off.

A parsing bug was fixed at the same time: shell comments inside the code blocks in the Operations chapter (# 1. Pull the new code) were being read as chapter headings, which silently replaced chapters 1-4 - Introduction, Roles & Permissions, Login, and the Store Keeper Manual - with fragments of a restore script. Ask "how do I log in?" today and you get §3.1, the login screen.

The role filter is unchanged and still applied before retrieval: a Store Keeper's context physically cannot contain an Admin chapter.

21.7 Serialised tool and equipment tracking

Two hammers share one item code, so scanning either label used to resolve to the same inventory line - the system could not tell you which one was in your hand or where the other had gone. Assets solves that.

Each physical item gets its own record, identified by its serial number. A scan now resolves three ways, and the order is deliberate:

What the label carries	What you get
A serial number or asset tag	That exact item, immediately - there is nothing to choose
Only the item code, and several units exist	A "which one is in your hand?" list to pick from

What the label carries	What you get
Only the item code, and no units are registered	The ordinary material view - the system does not invent an asset

Recording where something is. Open the item, tap Move, and either pick a rack or describe the place in words - not everything lives on a shelf, and "loaded on truck 4771" is a real answer. If the device can supply coordinates, they are captured alongside the update, never in front of it: a declined permission or a yard with no signal still records the move, with the coordinates simply absent. A recorded position becomes a map link on the item.

Status covers both what condition a tool is in - working, not in use, under repair - and where it sits in the stores workflow. The full history of every move is kept and is never overwritten, so "where has this been" is a question with an answer.

> Coordinate capture needs a secure connection. Over plain HTTP the browser > refuses to supply a position - the move still saves, without coordinates. If > your team is on the installed app or the company web address, this is already > the case.

> Your spreadsheet seeds this; the app owns it. A location typed in the > Consumption Log creates the asset on the next sync. From then on, what somebody > recorded in the app wins: a re-sync will not move a tool, revert its status, or > erase a captured position.

21.8 The warehouse rack locator

"Which rack is this in?" now has an answer. Search by material name, by code, or by scanning a label, and the system tells you the shelf to walk to - ordered so that the first result is where to go first, even when a material is stocked in three places.

- A material you stock but nobody has located comes back marked not located, rather than being left out. Silence would read as "we do not stock it", which sends somebody out to buy what is already on a shelf.
- Scanning the rack itself answers the reverse question - everything that is meant to be on that shelf. That turns a stock count from a hunt into a checklist.
- The locator is built into the CmdK search, so it is two keystrokes from anywhere in the application.

21.9 The Material Estimator handles messy tank references

The Consumption Log records a tank reference typed by hand, and the same vessel appears written four different ways. The system now recognises those as one vessel automatically - that alone accounted for 58 of 103 rows.

Where a reference is genuinely ambiguous, it refuses to guess. TNK-091 exists on both TRAIN J and TRAIN K; either answer would look completely plausible in every report afterwards, so those rows wait in a queue for someone who knows. Resolve the reference once on the Actual Consumption tab and every row waiting on it is tagged.

21.10 For administrators - local tooling

Two maintenance scripts ship with the system, documented in the administrator runbook: one puts the database and the public connection to sleep and wakes them again for a laptop that is not always on, and one takes and restores database snapshots, with an option to run nightly at 02:00.

> The previous nightly backup had been failing silently for 25 consecutive > nights - it pointed at a script that a repository reorganisation had moved. > Anyone relying on it had no local backups at all. The replacement is verified > by actually restoring into a scratch database on every change, which is the > only check that means anything.

21.11 Downloaded files now protect you from your own spreadsheet

If you type something that begins with an equals sign, a plus, a minus or an at-sign into a text field - a remark, a note, a reason - you may notice a small apostrophe in front of it when you open the downloaded Excel or CSV file.

That is deliberate, and nothing has been lost.

Spreadsheet programs treat a value beginning with one of those symbols as a calculation to run, not as words to show. That is fine when you write the formula yourself. It is not fine when the text arrived from somewhere else: a note typed at one site, exported by a manager and opened on their computer, would be run by their spreadsheet without anyone choosing to. The apostrophe tells the spreadsheet to show the text exactly as it was written instead.

What this means in practice:

- Numbers are never touched. Quantities, prices, areas and every total in every report stay real numbers, and every total still adds up.
- A plain number that happens to be negative, such as -5, is left exactly as it is - it is a number, not a calculation.
- Ordinary text - material names, supplier names, ordinary notes - is untouched.
- If you want the plain text back in your own copy, delete the apostrophe in that cell.

You are most likely to see this on remarks and reason fields, because those are the places people type freely. It applies to every download the system produces: reports, master-data exports and the Material Estimator workbooks alike.

Document end

This manual covers every page, tab, button, table and field in the General Industries Hub: the site workspaces (sections 4 to 7), the Reports module (section 8), the Logistics and Warehouse portals and the procurement chain (sections 14 to 16), hosting and operations (section 17), the Material Estimator (section 18), Man-Hours and Labor Tracking (section 19), the Auditor role (section 20), and the asset-tracking and warehouse-locator features added in 2026-08 (section 21).

For day-to-day operating procedure across all roles, see the Standard Operating Procedure document, which is shorter and organised by task rather than by screen.

Printed and PDF copies

Every role has its own booklet containing only the chapters that apply to them - a Store Keeper's booklet does not include the Logistics portal, and is a fraction of the length. Admin Portal -> Settings offers each of them as a download, and the full manual alongside.

If you are handing the system over to a new team, print the role booklets rather than this complete document. People read twenty relevant pages and skip two hundred.